

Trade Models, Trade Elasticities, and the Gains from Trade^{*}

Ina Simonovska[†]

Michael E. Waugh[‡]

UC Davis, NBER, CEPR

FRB Minneapolis, NBER

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Abstract

We consider five models of international trade that belong to the class of models in which the trade elasticity and domestic trade share are sufficient statistics for the welfare cost of autarky. Three are fixed-variety models featuring increasing micro-level margins of adjustment: versions of the Armington model, the Ricardian model, and its variable-markup counterpart. Two are endogenous-variety models featuring monopolistic competition: versions of the Krugman model and the Melitz model. For the three fixed-variety models, we show theoretically that, although they imply the same mapping from the trade elasticity to welfare, they imply different mappings from the trade elasticity to observed order statistics of price gaps of identical goods across countries. This result implies that choosing the parameter that governs the elasticity of trade to match the same order statistics in the data differentially impacts each model's measured welfare cost of autarky. We quantify these differences for all five models.

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[†]inasimonovska@ucdavis.edu

[‡]michael.e.waugh@gmail.com

1 Introduction

Trade barriers shape trade flows and prices of tradable goods across countries. In an important class of trade models, the trade elasticity and the share of expenditure on domestic goods are sufficient statistics to measure the welfare cost of autarky (Arkolakis et al., 2012) (henceforth ACR). A common practice in the literature is to apply a single estimate of the trade elasticity across all models in this class. We show that this practice can be misleading: although these models imply the same mapping from the trade elasticity to welfare, they imply different mappings from the trade elasticity to observed micro-level price variation. Choosing the parameter that governs the elasticity of trade to match the same price data therefore yields systematically different trade elasticities and welfare implications across models.

Our analysis covers five canonical models of trade. Three feature a fixed set of varieties: Anderson (1979) (henceforth Armington), Eaton and Kortum (2002) (henceforth EK), and Bernard et al. (2003) (henceforth BEJK). Two feature endogenous varieties under monopolistic competition: Krugman (1980) (henceforth Krugman) and Melitz (2003) (henceforth Melitz). These models are nested in the sense that each adds a micro-level margin of adjustment. Armington features an intensive margin only. EK adds an extensive margin: reductions in trade costs lead to additional goods being traded. BEJK further adds variable markups. Among the monopolistic competition models, Krugman features only an intensive margin, while Melitz adds firm selection.

Our theoretical analysis focuses on the three fixed-variety models. To understand how micro-level margins shape the mapping between the trade elasticity and price data, we examine the distribution of price gaps of identical goods across countries in a symmetric, two-country version of each model. For a fixed trade elasticity, we show that the price gap distribution in the Armington model first-order stochastically dominates that in EK, which further dominates the one in BEJK. The ranking arises because each additional margin compresses price dispersion. The active extensive margin in EK places positive mass on non-traded goods whose price gaps lie strictly below the trade cost. Variable markups in BEJK further compress price gaps because markups are negatively correlated with costs. These rankings carry over to the expected maximal price gap, or the largest order statistic, from a sample of goods.

The stochastic dominance result has two implications. First, it provides a basis for testing which model best describes the micro-level price patterns found in the data. Second, it implies that choosing the parameter that governs the trade elasticity to match the same order statistic of price gaps in the data will yield different trade elasticities across the three fixed-

variety models. Because the Armington distribution places the most mass on large price gaps, its trade elasticity parameter must be the highest to generate the same observed order statistic. BEJK, which compresses price gaps the most, requires the lowest. Combined with ACR’s sufficient statistic formula, this implies that models with more micro-level margins yield higher welfare gains from trade when disciplined by the same price data.

We quantify the importance of these theoretical results by choosing the parameter that governs the trade elasticity in the multi-country, asymmetric version of each model to match moments from the distribution of bilateral price gaps. The estimation approach builds on the theory described above and on Simonovska and Waugh (2014), who focus on the specifics of the EK model. The procedure has two steps. First, we estimate the structural parameters necessary to simulate micro-level prices from each model using bilateral trade-flow data, which ensures all models generate identical aggregate trade predictions. Second, given these parameter estimates, we choose the parameter that governs the trade elasticity in each model to minimize the distance between model-implied and data moments about order statistics of bilateral price gaps.

Using price and trade-flow data for the years 2004, 2011, and 2017 for the 30 largest countries in terms of gross output, we obtain the following results for the fixed-variety models. Across all specifications, the parameter governing the trade elasticity is systematically and significantly lower for BEJK relative to EK, and for EK relative to Armington. On average across years, the estimate for the Armington model is approximately 5.1, for EK approximately 4.1, and for BEJK approximately 3.3 (at a baseline CES elasticity of substitution of 2.5). Thus, the EK estimate is roughly 20 percent lower than Armington, and BEJK is a further 20 percent lower than EK. Through the lens of the Arkolakis et al. (2012) sufficient statistic formula, these differences translate directly into welfare: the welfare cost of autarky in BEJK is approximately 55 percent higher than in the Armington model. Importantly, these estimates are stable across the three time periods we examine.

The BEJK estimates are sensitive to the CES elasticity of substitution, which governs the monopoly markup bound and hence the degree of price compression from variable markups. At an elasticity of substitution of 1.5 (large markups), the BEJK estimate falls to approximately 2.8, while at a value of 3.0 (small markups, closer to EK), it rises to approximately 3.4. The range across these specifications is economically meaningful: it implies a variation in the welfare cost of autarky of roughly 20 percent within the BEJK model alone.

We extend the quantitative analysis to the two endogenous variety models. While the theoretical stochastic dominance results apply to the fixed-variety models, the same estimation procedure can be used to choose the trade elasticity parameter in Krugman and Melitz. In the baseline specification—where entry costs are proportional to country size so

that country size does not affect the gravity equation—the Melitz estimate is approximately 3.4 on average, roughly 34 percent below the Krugman estimate of approximately 5.1. The presence of productivity-based firm selection in Melitz compresses price gaps relative to Krugman, paralleling the effect of the extensive margin in EK relative to Armington.

We also examine an alternative specification of the Krugman model in which fixed entry costs are constant across countries, so that larger countries have more firms. This introduces a market-size effect into the gravity equation and the price index. Under this specification, the Krugman estimate falls substantially to approximately 3.7 on average because the market-size channel accounts for part of the trade variation that is otherwise attributed to the trade elasticity parameter. This result highlights the fact that the Krugman model that features a home market effect encompasses an extra margin of adjustment, akin to the Melitz model.

A natural question is why we focus on micro-level price variation rather than alternative approaches. First, all models make concrete and distinct predictions about price variation; in contrast, moments about the firm or sales size distribution are indeterminate in the Armington and EK models. Second, we document that micro-level price data are informative about trade barriers: price gaps correlate strongly with bilateral distance, tariff levels, and the presence of shared borders. Third, micro-level price variation speaks directly to the economic mechanisms at work in each model. The active extensive margin in EK or Melitz and variable markups in BEJK all manifest themselves in different patterns of price gaps. Thus, to learn about the importance of these margins, one should examine data that these margins can speak to. We make some preliminary steps in this direction by finding that the Armington, Krugman, and BEJK models do not describe well (in a formal statistical sense) the micro-level price patterns found in the data.

An important feature of our analysis is that we restrict the exercise to models that fall within the ACR class. Therefore, our approach is distinct from exploring deviations from CES preferences, unbounded Pareto-Melitz productivity distributions, trade dynamics, non-iceberg trade costs, etc. These deviations would constitute sources of differences in the welfare gains from trade.

Finally, a closely related paper is Melitz and Redding (2015), who examine the same class of endogenous variety models that we consider and they find the same welfare ranking between the Krugman and Melitz models. Like us, they place constraints on the comparison across models. Their approach holds common structural parameters (notably the elasticity of substitution) fixed and then shows, via a revealed preference argument, that the additional margin of adjustment in the Melitz model generates higher welfare gains than Krugman. We instead constrain models to match the same moments in the data—specifically,

variation in price gaps and bilateral trade shares—and show that the Melitz model yields larger gains from trade because doing so requires a lower trade elasticity parameter.

2 Trade Barriers and Prices

To motivate our estimation approach, which uses micro-price data, we describe our dataset and we document key information that price data reveal about cross-country trade barriers.

Our sample contains the thirty largest countries in the world (in terms of absorption). The price data at the basic-heading-level are obtained from the [International Comparison Programme](#) (ICP), made available by the World Bank and tasked by the United Nations Statistical Commission, and constitute the price observations underlying the computations of the Purchasing Power Parity (PPP) adjustment to national income statistics by the [Penn World Tables](#). This dataset represents the most comprehensive cross-country price comparison project for goods and services purchased via traditional means, i.e. in person. The dataset has been employed in a number of empirical studies. For example, Sposi (2015) uses the ICP price data to study the relationship between prices of tradable goods and per-capita income levels, Mutreja et al. (2018) examine prices of capital goods across countries, while Bradford (2003) and Bradford and Lawrence (2004) measure the degree of fragmentation, or the level of trade barriers, among OECD countries. Eaton and Kortum (2002) use a similar dataset for the year 1990 and Simonovska and Waugh (2014) use the data for year 2004 in order to estimate the parameter that governs the trade elasticity in the EK model.

The ICP periodically collects price data on goods (and services) with identical characteristics in retail locations in the participating countries. We use data from three consecutive cycles: 2004, 2011 and 2017.¹ The basic heading represents a narrowly defined group of goods for which expenditure data are available (ex. rice). The dataset typically contains a total of 129 basic headings, and we reduce the sample to 62-71 categories that correspond to manufacturing products, depending on the year (Table 1). Table 12 in Appendix D lists all the basic headings used in the analysis for year 2017. The list is similar in earlier years.

The ICP provides a common list of “representative” goods whose prices are to be randomly sampled in each country over a certain period of time. A good is representative of a country if it comprises a significant share of a typical consumer’s bundle there. Thus, the ICP samples the prices of a common basket of goods across countries, where the goods have been pre-selected due to their highly informative content for the purpose of international comparisons.

¹The ICP Methodological Handbook is available [here](#). Data from the post-pandemic collection round became recently available.

Table 1: Summary Statistics: $d_{ni,t}^d$ (Max Price Gap)

Year	N	Basic Headings	Mean	Median	Std	[Min, Max]
All	2604	—	0.938	0.855	0.418	[0.185, 3.128]
2004	866	62	0.924	0.898	0.321	[0.252, 2.268]
2011	868	71	0.980	0.829	0.514	[0.185, 3.128]
2017	870	64	0.911	0.835	0.394	[0.206, 2.258]

Below we document that ICP price data are highly informative of trade barriers. Specifically, for each basic heading and pair of countries, we compute the relative price, and then, for each country-pair, we examine order statistics from the relative price gap distribution across all basic headings. Moreover, for each country-pair, we deflate the relative price gap by the relative country-level price indices, which constitute the geometric means of prices across all basic headings within each country. This procedure eliminates country-level price distortions and allows us to hone in on the cross-country component in prices.

Focusing on country pairs with positive trade flows, we examine how moments from the price gap distribution correlate with standard variables that explain bilateral trade flows, most notably gravity controls and bilateral tariffs. We source trade flows from UN Comtrade for 2004 (Simonovska and Waugh, 2014) and BACI (Gaulier and Zignago, 2010) for 2011 and 2017 at the HS6-level, and we aggregate them to the bilateral country-level. Manufacturing trade flows include HS 2-digit codes from 28 to 96. We compute bilateral import tariffs for the manufacturing sector as a simple average of sector-specific tariffs across ISIC 2-digit sectors ranging from 10 to 33 using the Global Tariff Database (Teti, 2024). Gravity controls are from the Dynamic Gravity Dataset (Gurevich, 2018) and include distance, as well as an indicator variable for contiguity.

We begin by examining the price gap moment d_{ni} , which is the key data input for identifying the trade elasticity (Eaton and Kortum, 2002; Simonovska and Waugh, 2014). It measures bilateral price discrepancies across countries using ICP price level data on traded goods categories. For each country pair (n, i) and year t , the price gap is constructed as

$$d_{ni,t}^d = \max_k \left(\log p_{n,t}^k - \log p_{i,t}^k \right) - \text{mean}_k \left(\log p_{n,t}^k - \log p_{i,t}^k \right), \quad (1)$$

where k indexes traded goods categories (ICP basic headings). The alternative measure $d_{ni2,t}^d$ replaces the maximum with the 85th percentile and can be found in Appendix D.

Table 1 reports summary statistics for the price gap measure. The mean $d_{ni,t}^d$ is close to one across all years, meaning that the maximum log-price deviation within a country pair typically exceeds the mean deviation by about one log point. The 2011 sample exhibits the

Table 2: Correlations: $d_{ni,t}^d$ and Gravity Variables

Year	N	$\rho(d_{ni}, \log \text{dist})$	p	$\rho(d_{ni}, \text{border})$	p	$\rho(d_{ni}, \log(1 + \tau))$
All	2604	0.422	0.000	-0.183	0.000	0.338
2004	866	0.278	0.000	-0.108	0.001	0.261
2011	868	0.472	0.000	-0.196	0.000	0.409
2017	870	0.501	0.000	-0.240	0.000	0.450

widest range and highest dispersion.

We next examine Pearson correlations between the log price gap moments and three standard gravity variables. The expected signs are: positive for distance (more distant pairs face larger trade costs and hence larger price gaps), negative for shared borders (contiguity reduces trade costs and thus price gaps), and positive for tariffs (higher barriers generate larger price gaps). Tables 2 and 14 report the results.

All correlations are highly significant and carry the expected sign in every year and for both measures. For $d_{ni,t}^d$, distance is the strongest correlate (between 0.28 and 0.50), followed by tariffs (0.26 to 0.41), with border effects more modest (-0.11 to -0.20).

To assess the joint explanatory power of gravity variables, we estimate OLS regressions of the form

$$d_{ni,t}^d = \alpha_t + \beta_1 \cdot \text{border}_{ni} + \beta_2 \cdot \log \text{dist}_{ni} + \beta_3 \cdot \log(1 + 0.01 \cdot \text{tariff}_{ni}) + \varepsilon_{ni}, \quad (2)$$

with year fixed effects. Table 3 reports the results.

Distance and tariffs are highly significant in both regressions. A 10 percent increase in bilateral distance raises $d_{ni,t}^d$ by approximately 0.014. Tariffs are also highly significant, with coefficients of 1.97 and 0.39. The border indicator is insignificant in both regressions, suggesting that contiguity effects are largely absorbed by distance.

This systematic relationship between price gaps and gravity variables suggests that price gaps they contain valuable information about trade barriers. In the next section, we formalize the information content of price gaps on the underlying model parameters through the lens of canonical models of international trade.

3 Fixed Variety Models of Trade

Below we outline three canonical models of trade and we formalize the relationship between trade flows, micro-level prices and trade elasticities. Throughout the paper, we use the definition of a “good” j , which is often referred to as a “variety” by the trade literature.

Table 3: OLS Regression: $d_{ni,t}^d$ and Gravity Variables

Variable	Coef.	Std. Err.	t	p	95% CI
Intercept	-0.235	0.069	-3.40	0.001	[-0.370, -0.099]
Border	-0.032	0.034	-0.96	0.339	[-0.098, 0.034]
log(dist)	0.135	0.009	15.11	0.000	[0.117, 0.152]
log(1 + τ)	1.973	0.201	9.81	0.000	[1.579, 2.367]
$R^2 = 0.212, \quad N = 2,604, \quad F = 139.9$					

3.1 Armington

The simplest model of international trade that yields a gravity equation is the Armington model outlined in Anderson and van Wincoop (2003). The framework features N countries populated by consumers with constant elasticity of substitution (CES) preferences and tradable goods that are differentiated by the country of origin. Perfect competition among producers and product differentiation by origin imply that each good is purchased in each destination at a price that equals the marginal cost of production and delivery of the good there.

We consider a more empirically-relevant version of the Armington model that features product differentiation within and across countries.² Throughout the paper, let i denote the source country and n the destination. We allow each country to produce an exogenously given measure of tradable goods equal to $1/N$. We assume that products are differentiated. Within each country n , there is a measure of consumers L_n . Each consumer has one unit of time supplied inelastically in the domestic labor market and enjoys the consumption of a CES bundle of final tradable goods with elasticity of substitution $\rho_{\text{ARM}} > 1$,³

$$U_n = \left[\sum_{i=1}^N \int_0^{1/N} x_{ni}(j)^{\frac{\rho_{\text{ARM}}-1}{\rho_{\text{ARM}}}} dj \right]^{\frac{\rho_{\text{ARM}}}{\rho_{\text{ARM}}-1}}.$$

To produce quantity $x_{ni}(j)$ of good j in country i , a firm employs labor using a linear production function with productivity $T_i^{1/\theta_{\text{ARM}}}$, where T_i is a country-specific technology parameter and $\theta_{\text{ARM}} > 0$ is a scaling parameter that is common across countries. The perfectly competitive firm from country i incurs a marginal cost to produce good j of $w_i/T_i^{1/\theta_{\text{ARM}}}$, where

²This model yields identical aggregate predictions to the simple Armington model, but it accommodates a greater number of goods than countries, which is an observation that holds true in the data.

³ $j \in [0, 1/N]$ is an index for an individual good. Since goods are differentiated by country of origin, a good should be denoted by j, i ; however, we suppress the i notation for brevity as prices and quantities are denoted by origin i and destination n .

w_i is the wage rate in the economy. Shipping the good to a destination n further requires a per-unit iceberg trade cost of $\tau_{ni} > 1$ for $n \neq i$, with $\tau_{ii} = 1$. We assume that cross-border arbitrage forces effective geographic barriers to obey the triangle inequality: For any three countries i, k, n , $\tau_{ni} \leq \tau_{nk}\tau_{ki}$. We maintain this assumption about the nature of iceberg trade costs in all the models that we describe below.

Perfect competition forces the price of good j from country i to destination n to be equal to the marginal cost of production and delivery

$$p_{ni}(j) = \frac{\tau_{ni}w_i}{T_i^{1/\theta_{\text{ARM}}}}.$$

Since goods are differentiated, consumers in destination n buy all products from all sources and pay $p_{ni}(j)$ for good j from i . Hence, all tradable goods are traded among the N markets and consumers buy a unit measure of goods.

3.2 Eaton and Kortum (2002)

We now outline the environment of the multi-country Ricardian model of trade introduced by Eaton and Kortum (2002)—EK. As in the Armington model, there is a continuum of tradable goods indexed by $j \in [0, 1]$. Preferences are represented by the following utility function

$$V_n = \left[\int_0^1 x_n(j)^{\frac{\rho_{\text{EK}}-1}{\rho_{\text{EK}}}} dj \right]^{\frac{\rho_{\text{EK}}}{\rho_{\text{EK}}-1}},$$

where $\rho_{\text{EK}} > 1$. To produce quantity $x_i(j)$ in country i , a firm employs labor using a linear production function with productivity $z_i(j)$. Unlike the Armington model, country i 's productivity is the realization of a random variable (drawn independently for each j) from its country-specific Fréchet probability distribution

$$F_{EK,i}(z_i) = \exp(-T_i z_i^{-\theta_{\text{EK}}}).$$

The country-specific parameter $T_i > 0$ governs the location of the distribution; higher values of it imply that a high productivity draw for any good j is more likely. The parameter $\theta_{\text{EK}} > \rho_{\text{EK}} - 1$ is common across countries and, if higher, it generates less variability in productivity across goods.

Having drawn a particular productivity level, a perfectly competitive firm from country i incurs a marginal cost to produce good j of $w_i/z_i(j)$. Perfect competition forces the price

of good j from country i to destination n to be equal to the marginal cost of production and delivery

$$p_{ni}(j) = \frac{\tau_{ni}w_i}{z_i(j)}.$$

So, consumers in destination n would pay $p_{ni}(j)$, should they decide to buy good j from i .

Consumers purchase good j from the low-cost supplier; thus, the actual price consumers in n pay for good j is the minimum price across all sources k

$$p_n(j) = \min_{k=1,\dots,N} \left\{ p_{nk}(j) \right\}.$$

Hence, the EK framework introduces endogenous tradability into the Armington model outlined above. In particular, countries export only a subset of the unit measure of tradable goods for which they are the most efficient suppliers.

3.3 Bernard, Eaton, Jensen, and Kortum (2003)

Bernard et al. (2003)—BEJK—introduce Bertrand competition into EK's model. The most important implication from this extension is that individual good prices differ from the EK model.

Let $c_{kni}(j) \equiv \tau_{ni}w_i/z_{ki}(j)$ be the cost that the k -th most efficient producer of good j in country i faces in order to deliver a unit of the good to destination n . With Bertrand competition, as with perfect competition, the low-cost supplier of each good serves the market. For good j in market n , this supplier has the following cost $c_{1n}(j) = \min_i \{c_{1ni}(j)\}$. This supplier is constrained not to charge more than the second-lowest cost of supplying the market, which is $c_{2n} = \min \{c_{2ni^*}(j), \min_{i \neq i^*} \{c_{1ni}(j)\}\}$, where i^* satisfies $c_{1ni^*}(j) = c_{1n}(j)$. Hence, the price of good j in market n is

$$p_n(j) = \min \{c_{2n}(j), \bar{m}c_{1n}(j)\},$$

where $\bar{m} = \rho_{EK}/(\rho_{EK} - 1)$ is the Dixit-Stiglitz constant mark-up.

Finally, for each country i , productivity, $z_{ki}(j)$ for $k = 1, 2$ is drawn from

$$F_{BEJK,i}(z_1, z_2) = \left[1 + T_i(z_2^{-\theta_{BEJK}} - z_1^{-\theta_{BEJK}}) \right] \exp\left(-T_i z_2^{-\theta_{BEJK}}\right),$$

where $\theta_{BEJK} > \rho_{EK} - 1$. Hence, the BEJK model features a key additional component relative to the EK framework: the existence of variable mark-ups. In particular, the most efficient

suppliers in this model enjoy the highest mark-ups.

3.4 Trade Flows, Aggregate Prices, and Welfare

The models described above produce identical aggregate outcomes, even though they feature different micro-level behavior. In particular, under certain parametric assumptions, the models yield the same expressions for trade flows, price indices (up to a constant scalar), and welfare gains from trade. Proposition 1 summarizes the result.

Proposition 1. Assume $\theta_{ARM} = \rho_{ARM} - 1$. Given the functional forms for productivity $F_{EK,i}(\cdot)$, $F_{BEJK,i}(\cdot)$ for all $i = 1, \dots, N$,

- a. The share of expenditures that n spends on goods from i , X_{ni}/X_n , predicted by the models is

$$\frac{X_{ni}}{X_n} = \frac{T_i(\tau_{ni}w_i)^{-\theta_M}}{\sum_{k=1}^N T_k(\tau_{nk}w_k)^{-\theta_M}}. \quad (3)$$

- b. The CES exact price index for destination n , P_n , predicted by the models is

$$P_n \propto \Phi_n^{-\frac{1}{\theta_M}}, \quad \text{where} \quad \Phi_n = \sum_{k=1}^N T_k(\tau_{nk}w_k)^{-\theta_M}. \quad (4)$$

- c. The percentage compensation that a representative consumer in n requires to move between two trading equilibria predicted by the models is

$$\frac{P'_n}{P_n} - 1 = 1 - \left(\frac{X'_{nn}/X'_n}{X_{nn}/X_n} \right)^{\frac{1}{\theta_M}}, \quad (5)$$

where $M = ARM, EK, BEJK$

We prove a. and b. for the Armington model in the Appendix. The results for the EK and BEJK models are derived in the respective papers. The proof of part c. can be found in Arkolakis et al. (2012). Across the models, the welfare gains from trade are essentially captured by changes in the CES price index that a representative consumer faces. Using the objects from the Proposition above, it is easy to relate the price indices to trade shares and the parameter θ_M . In particular, expressions (3) and (4) allow us to relate trade shares to trade costs and the price indices of each trading partner via the following equation

$$\frac{X_{ni}/X_n}{X_{ii}/X_i} = \frac{\Phi_i}{\Phi_n} \tau_{ni}^{-\theta_M} = \left(\frac{P_i \tau_{ni}}{P_n} \right)^{-\theta_M}, \quad (6)$$

where $\frac{X_{ii}}{X_i}$ is country i 's expenditure share on goods from country i , or its home trade share. The welfare equation follows trivially from this expression.

3.5 Trade Elasticities Across Models

The key parameter determining trade flows (equation (6)) and welfare (equation (5)) is θ_m . To see the parameter's importance for trade flows, take logs of equation (6) yielding

$$\log\left(\frac{X_{ni}/X_n}{X_{ii}/X_i}\right) = -\theta_m [\log(\tau_{ni}) + \log(P_i) - \log(P_n)]. \quad (7)$$

As this expression makes clear, θ_m controls how a change in the bilateral trade costs, τ_{ni} , will change bilateral trade between two countries predicted by each model.

To see the parameter's importance for welfare, it is easy to demonstrate that (5) implies that θ_m represents the inverse of the elasticity of welfare with respect to domestic expenditure shares for each model

$$\log(P_n) = -\frac{1}{\theta_m} \log\left(\frac{X_{nn}}{X_n}\right). \quad (8)$$

Hence, decreasing the domestic expenditure share by one percent generates a $(1/\theta_m)/100$ -percent increase in consumer welfare.

Expression (8) makes clear that, in order to measure the impact of trade policy on welfare, it is sufficient to obtain data on realized domestic expenditures and an estimate of the elasticity of trade. Furthermore, expression (7) makes clear that the challenge with estimating the trade elasticity is that one must *separately* disentangle θ_m from trade costs, which are not observed. The leading approach in the existing literature is to instrument trade costs with observed variables, most notably geographic indicators (such as distance and contiguity) and policy variables (tariffs) (see Boehm et al. (2023) for a detailed discussion and quantification of the numerous biases that arise when using tariff data for identification).

We opt for a different approach—one that relies on relative prices across goods and across countries to infer information about trade elasticities. Our motivation draws from the empirical observation in Section 2 that order statistics from the distribution of relative prices are systematically correlated with distance, contiguity and tariffs, and therefore contain information about trade barriers.

To formalize the argument, we theoretically show that micro-level cross-country price variation contains information about the parameter that governs the trade elasticity in each model. More importantly, different models represent different mappings between price gaps

and trade elasticity parameters, which implies that, in order to match the observed price variation in the data, different models require different trade elasticity parameters.

4 Price Gaps and Trade Elasticities

This section shows that different trade models have different implications for the distribution of relative price gaps. This implies that, given data on trade flows and micro-level prices, different models require different values for the parameter that governs the trade elasticity in each model; thus the measured welfare gains from trade in each model are different.

To illustrate the argument theoretically, we focus on a symmetric, two country world with the countries denoted as home (h) and foreign (f). In a symmetric two-country world, the only two parameters that govern trade flows (as well as prices) in the models above are θ_m and τ .⁴ Throughout this section, we assume that θ_m and τ are constrained to exactly fit a given value for the bilateral trade share. This implies that if $\theta > \theta'$, then $\tau < \tau'$.

Given the constraint that all models must imply the same bilateral trade flows, we then make the following argument. First, we describe properties of the distribution of relative prices at the micro-level. Second, we order these distributions (across models and for different θ s) by first-order stochastic dominance and, in turn, by the largest order statistic or maximal price gap in a finite sample. This then allows us to order the implied θ_m s across models given the observed largest order statistic from a sample of data.

4.1 Ordering Price Distributions and the Largest Order Statistic Across Models

In this section, we focus on deviations from the law of one price at the micro level, i.e. for an individual variety. Specifically, we focus on the logged price gap between the home and the foreign country:

Definition 1. *For any micro-level good ℓ , define the random variable that equals the logged price gap between the home and the foreign country as $\log \tilde{P}_{h,\ell} \equiv \log P_{h,\ell} - \log P_{f,\ell}$.*

Throughout, we drop the subscript ℓ for brevity, and we only use it when it is necessary to distinguish prices across goods.

⁴To see this, from the first equality in expression (6) and the definition of Φ_n in (4), notice that, given trade share data and a value for θ_m , $\log \tau$ can be computed immediately in a symmetric two-country world.

The distribution of price gaps is an important object of interest. Given a model $\mathcal{M}$, denote the cumulative density function of log price gaps, conditional on the price gap being positive as:

$$\forall \text{ models } \mathcal{M}, \quad G_{\mathcal{M}}(\log \tilde{p}_h, \theta) = \begin{cases} 0 & : \log \tilde{p}_h < 0 \\ \in [0, 1] & : 0 \leq \log \tilde{p}_h \leq \log \tau \\ 1 & : \log \tau < \log \tilde{p}_h. \end{cases} \quad (9)$$

Note that we index the density by parameter θ . This indexing is sufficient because other parameters are either the same across models (e.g. the technology parameters), or are determined by the value θ , e.g. θ determines τ so aggregate trade is held constant.

An important property of this density is that all price gaps lie below $\log \tau$. The reason is the following: suppose that $\log P_{h,\ell} - \log P_{f,\ell} > \log \tau$, then an arbitrage opportunity exists as an agent could import good ℓ from the foreign country at a lower price. Thus, this inequality places an upper bound on the possible observable price gaps. This property is important because the support of the density depends on θ . Different θ s imply different τ s because aggregate trade constrains the values that these parameters can take. Thus, this property allows us to make conclusions as to how parts of the density shift in a model with parameter θ relative to the model with parameter θ' .

How the density shifts as θ changes at any price gap in certain models is analytically intractable (though numerically verifiable). Thus, we assert that the following regularity conditions holds on the price gap distribution.

Regularity Condition 1. (No Crossing.) *Let the density $G_{\mathcal{M}}(\log \tilde{p}_h, \theta)$ satisfy the following no-crossing property. If $\theta' > \theta$, then for $\log \tilde{p}_h < \log \tau'$:*

$$\text{Prob}_{\mathcal{M}}(\log \tilde{P}_h < \log \tilde{p}_h, \theta) \leq \text{Prob}_{\mathcal{M}}(\log \tilde{P}_h < \log \tilde{p}_h, \theta'). \quad (10)$$

This property says the following: Fix a model $\mathcal{M}$. If we lower the elasticity from θ' to θ and constrain aggregate trade to stay the same, the probability of seeing a particular price gap cannot increase. So as the upper end of the support increases from $\log \tau'$ to $\log \tau$, it can not induce a “cross” in the density.

This property clearly holds in the Armington model. In the Armington model, all goods are traded and thus the probability mass of the price gaps lies completely at $\log \tau'$. If we lower the elasticity from θ' to θ , then all the probability mass shifts rightward to $\log \tau$. For the EK model, this holds as well (see online appendix of Simonovska and Waugh (2014)). For the BEJK model, analytically verifying this is not possible because a closed-form expres-

sion for the price gap distribution does not exist; we have, however, verified that this holds numerically for all relevant parameter values.

Below, we define the data features we will focus on. Specifically, the maximal price gap from a sample of price gaps:

Definition 2. Consider a finite, random sample of positive price gaps $\log \tilde{P}_{h,1}, \log \tilde{P}_{h,2}, \dots, \log \tilde{P}_{h,L}$ which are ordered from smallest to largest. The maximal price gap in the sample of L goods is $\log \tilde{P}_{h,L:L}$. Given a model $\mathcal{M}$, parameter θ , and density $G_{\mathcal{M}}(\log \tilde{p}_h, \theta)$, the expected maximal price gap or largest order statistic is $\mathbb{E}(\log \tilde{P}_{h,L:L}; \{\theta, \mathcal{M}\})$, where $\mathcal{M}$ indexes the model and the dependence on the parameter θ is noted.

Focusing on the maximal price gap has several appealing features. First, it has some history of thought as it has been used in Eaton and Kortum (2002) and Simonovska and Waugh (2014) in the estimation of θ . Second, order statistics encode much information about the underlying density. For example, one can show that a sequence of largest order statistics (that is for $L = 1, 2, \dots$) completely characterizes the density G (see, e.g., Arnold et al. (1992)). We will never have access to an infinite sequence of largest order statistics. However, by utilizing recurrence relationships between largest order statistics of different sample sizes (again see Arnold et al. (1992)), one can construct testable restrictions for each model.

Lemma 1 connects the expected maximal price gap with a stochastic dominance relationship in the density.

Lemma 1. If the price density, $G_{\mathcal{M}}(\log \tilde{p}_h, \theta)$, first-order stochastically dominates the price density $G_{\mathcal{M}'}(\log \tilde{p}_h, \theta')$, then $\mathbb{E}(\log \tilde{P}_{h,L:L}; \{\theta, \mathcal{M}\}) > \mathbb{E}(\log \tilde{P}_{h,L:L}; \{\theta', \mathcal{M}'\})$.

Proof: If a random variable first-order stochastically dominates another, the same ranking holds for the distribution of the variables' maxima, that is $G_{\mathcal{M}}(\log \tilde{p}_h, \theta)^L >_{fosd} G_{\mathcal{M}'}(\log \tilde{p}_h, \theta')^L$. This property implies that the expected maximal price gap under model $\mathcal{M}$ is larger than in model $\mathcal{M}'$. ■

The result here is that if one density dominates another density, this implies that the maximal price gap must be larger in the model that has the density which dominates. Application of Lemma 1 and the regularity condition above allows us to rank densities according to θ for a given model $\mathcal{M}$.

Lemma 2. For a given model, $\mathcal{M}$, the parameter θ indexes the price density by first-order stochastic dominance, and strictly indexes the expected maximal price gap. That is, if $\theta' > \theta$ then $G_{\mathcal{M}}(\log \tilde{p}_h, \theta) >_{fosd} G_{\mathcal{M}}(\log \tilde{p}_h, \theta')$ and $\mathbb{E}(\log \tilde{P}_{h,L:L}; \{\theta, \mathcal{M}\}) > \mathbb{E}(\log \tilde{P}_{h,L:L}; \{\theta', \mathcal{M}\})$.

Proof: First, recall that, when $\log \tau$ and θ are constrained to fit bilateral trade shares, $\theta' > \theta$ implies that $\tau' < \tau$. Then, $\tau' < \tau$ and Regularity Condition 1 implies that, for $\log \tilde{p}_h <$

$\log \tau'$, the cumulative probability in the model with θ is weakly less than the cumulative probability in the model with θ' . For $\log \tilde{p}_h = \log \tau'$, the cumulative probability in the model with θ' is equal to one, and because there is strictly positive probability mass in the region $\log \tilde{p}_h > \log \tau'$ for the model with θ , this implies a strict inequality in probability mass at $\log \tilde{p}_h = \log \tau'$. This implies first-order stochastic dominance. From Lemma 1, the ordering of the order statistics follows. ■

Lemma 2 allows us, for a given model, $\mathcal{M}$, an ability to strictly rank expected maximal price gaps by θ . One implication of this result is that the expected maximal price gap identifies the θ parameter. We exploit this result in Proposition 3. The second implication is that it assists us in showing that if all models have the same expected maximal price gap, then they must have different θ s. We state the result formally below:

Proposition 2. *Consider the models $G_{ARM}(\log \tilde{p}_h, \theta_{ARM})$, $G_{EK}(\log \tilde{p}_h, \theta_{EK})$, $G_{BEJK}(\log \tilde{p}_h, \theta_{BEJK})$. If the expected maximal price gaps are the same in all models:*

$$\mathbb{E}(\log \tilde{P}_{h,L:L}; \{\theta, \mathcal{M}\}) = \mathbb{E}(\log \tilde{P}_{h,L:L}; \{\theta', \mathcal{M}'\}), \quad \forall \text{ models } \mathcal{M},$$

then

$$\theta_{ARM} > \theta_{EK} \text{ and } \theta_{ARM} > \theta_{BEJK}.$$

Moreover, if $G_{EK}(\log \tilde{p}_h, \theta) >_{fosd} G_{BEJK}(\log \tilde{p}_h, \theta)$, then

$$\theta_{ARM} > \theta_{EK} > \theta_{BEJK}.$$

Proof: The proof proceeds by contradiction in the following two cases with the focus on the EK model. The same argument applies with respect to the BEJK model.

- Case #1. Suppose not and that $\theta_{ARM} = \theta_{EK} = \theta$. First, note that for the same θ ,

$$G_{ARM}(\log \tilde{p}_h, \theta) >_{fosd} G_{EK}(\log \tilde{p}_h, \theta), \quad (11)$$

because for any $\log \tilde{p}_h < \log \tau$, the probability mass in the Armington model is zero as all goods are traded and hence all positive price differences are equal to the trade friction. This is strictly less than in the EK model as for any $\log \tilde{p}_h < \log \tau$, the probability mass is greater than zero because there are non-traded goods. This argument implies the distribution of price gaps in the Armington model first-order stochastically dominates the distribution in the EK model. Application of Lemma 1 then implies the expected maximal price gaps should be ranked, which is a contradiction.

- Case #2. Suppose not and that $\theta_{EK} > \theta_{ARM}$. First, notice that this implies that $G_{ARM}(\log \tilde{p}_h, \theta_{ARM}) >_{f_{osd}} G_{ARM}(\log \tilde{p}_h, \theta_{EK})$ from Lemma 2. This then implies that,

$$\mathbb{E}(\log \tilde{P}_{h,L:L}; \{\theta_{ARM}, ARM\}) > \mathbb{E}(\log \tilde{P}_{h,L:L}; \{\theta_{EK}, ARM\}) > \mathbb{E}(\log \tilde{P}_{h,L:L}; \{\theta_{EK}, EK\})$$

and the last inequality follows from the observation made in Case #1 above, that for the same θ , the Armington price gap distribution strictly dominates the EK price gap distribution. This implication contradicts the assumption that the expected maximal price gap is the same in all models.

Finally note that the same exact arguments apply for the BEJK model, where some goods are non-traded as in the EK model. Thus, if the Armington model, EK model, and BEJK model, all have the same expected maximal price gap, then the elasticity in the Armington model must be strictly greater than the elasticity in EK or BEJK. ■

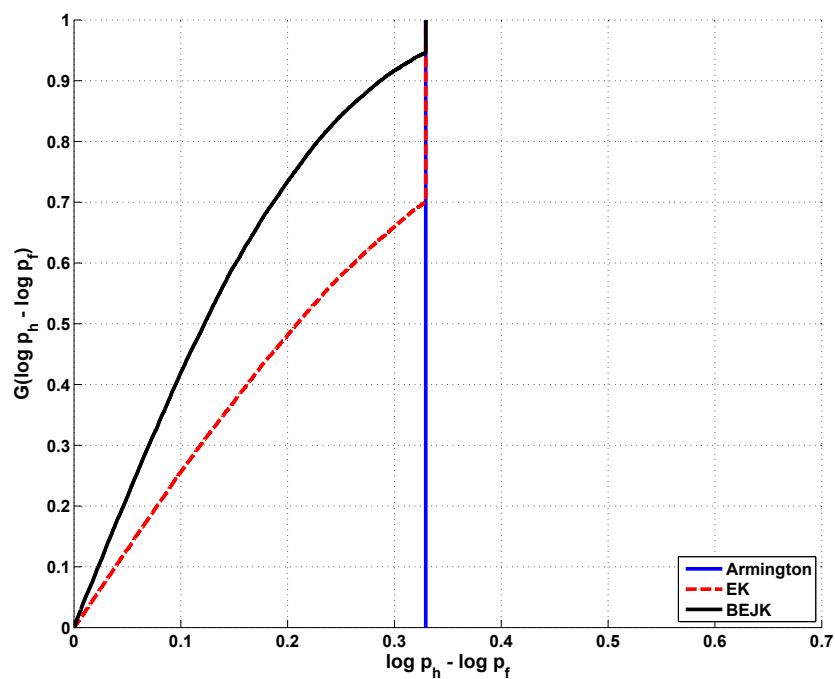
Figure 1 illustrates the intuition behind this result. The top panel plots the cumulative distribution function of log price gaps for the Armington (solid blue line), the EK (dashed red line) and the BEJK (solid black line) model—for the same θ .

In the Armington model, notice that all the mass lies on $\log \tau$. This is because all goods are traded, i.e. there is no active extensive margin in the Armington model. If the home country imports the good from the foreign country, then the price gap exactly reflects the trade friction. The distribution of price gaps in the EK model is different as it places a positive mass in the region between zero and the trade friction and hence the density in EK is stochastically dominated by that of Armington. This is because there are endogenously non-traded goods, i.e. there is an active extensive margin. In the EK model, there are instances where consumers in *both* countries find it cheaper to consume a variety from the local producer rather than importing the good. If a good is non-traded, the price difference across locations reflects differences in costs which are strictly less than the trade friction. This implies a positive mass in the region between zero and the trade friction.

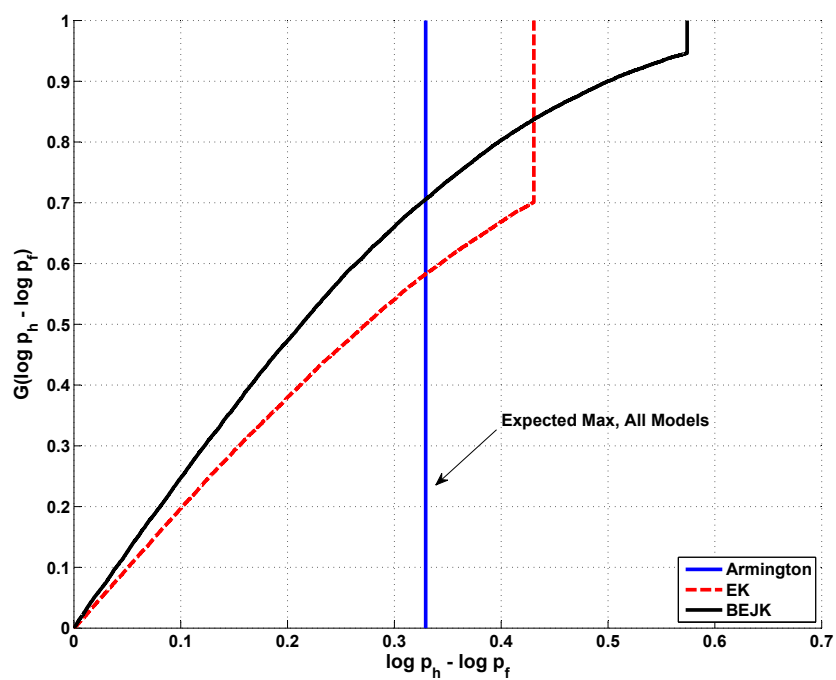
Proposition 2 says that because the price gap density in Armington dominates the density in EK, the only way for the expected maximal price gap to be the same is if the EK model has a lower θ . The bottom panel of Figure 1 illustrates this by plotting the densities when they have the same expected largest order-statistic. Here, the EK density is spread to the left with a larger upper bound to ensure that the largest order statistic is the same as in the Armington model.

What about the relationship between EK and BEJK? Generically, we can say the following: if the density in EK strictly stochastically dominates the density in BEJK and the largest order statistics are the same, then by logic of Proposition 2 this implies that $\theta_{EK} > \theta_{BEJK}$. A

Figure 1: Price Gap Distribution: Armington, EK, BEJK.



Same θ



Same Largest Order Statistic

stronger statement is not possible as it is difficult to theoretically construct the BEJK price gap distribution because of the presence of variable markups.⁵

The top panel in Figure 1 shows that for the same θ , the price gap density in EK dominates the density in BEJK. More generally we have verified that this relationship holds numerically for all relevant parameter values. And our estimates of θ in a multi-country setting with asymmetries always result in a ranking of $\theta_{EK} > \theta_{BEJK}$, under various parameter values for ρ . Furthermore, the bottom panel of Figure 1 verifies that, if the largest order statistic is the same in EK and BEJK, then the BEJK model must have a lower θ . This can be seen as the density in BEJK is spread even further to the left with a larger upper bound to ensure that the largest order statistic equals that in the EK and Armington models.

Variable markups and how they correlate with costs in the BEJK model explain why the price gap density in EK strictly dominates the density in BEJK. In the EK model, price gaps correspond to cost gaps and these cost gaps are identical in the two models. In BEJK, cost gaps do not correspond with price gaps as producers are able to price at a markup over marginal cost depending on other latent competitors.⁶ Thus, the price gap in BEJK reflects markups, cost differences (if the good is non-traded), and trade frictions (if the good is traded). In BEJK, markups are negatively correlated with marginal costs. This results in the BEJK distribution being more compressed than the EK distribution.⁷

Consider the following example to illustrate this point. Take a producer in the foreign country who has a very high productivity and is the lowest cost producer in her country of origin as well as the home country. This producer will likely charge a high markup (call it m') in the foreign country because she has a very low cost relative to her latent competitors as that is her country of origin. In contrast, she will likely charge a relatively lower markup (call it m'') in the home country because her cost advantage is eroded from the trade friction faced when exporting to the home country. This implies that the price gap, $\log \tilde{p}_h$, will be *less than* the trade friction even though the good is traded, i.e. the price gap equals $m'' - m' + \log \tau < \log \tau$. This observation implies that the BEJK distribution will have less mass around the log of the trade friction because of producers differentially marking up their products across locations.

⁵In the Appendix, we show that, in the case in which $\rho = 1$ and the Dixit-Stiglitz mark-up is infinite, the probability that the log price gap reaches the boundaries is strictly lower in the BEJK than in the EK model. This implies that the mass has to be distributed between these two end-points—meaning, in the non-traded good region. However, the exact shape of the distribution cannot be easily characterized.

⁶We exactly follow the treatment of latent competitors as in Bernard et al. (2003), which places the model within the ACR class of models. de Blas and Russ (2015) discuss the importance of the number of latent competitors in BEJK.

⁷A related property is that it is straightforward to show that the EK price distribution (not price gaps) is a mean preserving spread of the BEJK distribution.

4.2 Estimating θ s and the Gains from Trade

We now connect Proposition 2 with an econometrician's inference about θ given a sample of data (i.e. aggregate trade flows and a sample of micro-level prices) and discuss how this inference depends upon assumptions about the underlying model. Proposition 3 suggests a simple method of moments estimator and states that the estimated θ will (i) depend on the econometrician's assumptions about the underlying data generating process and (ii) will be ordered across models.

Proposition 3. *Consider the following estimator of θ that chooses $\hat{\theta}$ to minimize the distance between the observed largest order statistic $\log \tilde{P}_{h,L:L}$ and the expected largest order statistic $\mathbb{E}(\log \tilde{P}_{h,L:L}; \{\theta, \mathcal{M}\})$. That is,*

$$\hat{\theta}_{\mathcal{M}} = \arg \min_{\theta} h(\{\theta, \mathcal{M}\})' h(\{\theta, \mathcal{M}\})$$

$$\text{where } h(\{\theta, \mathcal{M}\}) = (\log \tilde{P}_{h,L:L} - \mathbb{E}(\log \tilde{P}_{h,L:L}; \{\theta, \mathcal{M}\})).$$

Then:

1. The estimate $\hat{\theta}_{ARM}$ is strictly greater than the estimate $\hat{\theta}_{EK}$ and $\hat{\theta}_{BEJK}$.
2. The estimate of the welfare cost of autarky in the Armington model is strictly less than the estimate of the welfare cost of autarky in EK and BEJK.
3. If the density in EK stochastically dominates the density in BEJK, then the estimate $\hat{\theta}_{EK} > \hat{\theta}_{BEJK}$ and estimate of the welfare cost of autarky in EK is strictly less than in BEJK.

Proof: The first statement follows from Lemma 2 which implies that the order statistic uniquely determines the θ for a given model and Proposition 2 which implies that across models, the θ must differ. The second statement follows from the fact that equation (8) is the same across models and that, by construction, all models have the same predictions for aggregate trade flows, but different θ . ■

A couple of comments are in order about this result. First, the estimator is simple. Our estimation strategy in Section 5 builds off the intuition from this result. Moreover, we include alternative moment conditions to enrich the estimation, to be able to test over-identifying restrictions, and to exploit empirical regularities we see in the data. We discuss these moment conditions in Section 5.

Second, compare Proposition 3 to the discussion in Section 6 in Arkolakis et al. (2012). In this section, they propose an alternative estimator of the trade elasticity that delivers

a common estimate of θ across models. Their estimator uses aggregate trade flows and some measure of trade frictions (ex. tariffs) and the common connection between aggregate trade flows (i.e. equation (7)) that all these models share. The final piece of their argument is to *assume* that the orthogonality condition that the estimate of θ is based on is model independent. In other words, their analog to the expectation of the function $h(\{\theta, \mathcal{M}\})$ is assumed not to depend on the model $\mathcal{M}$. Thus, the estimate of θ as well as the gains from trade will be the same across models.

The key distinction between our argument and that of Arkolakis et al. (2012) is that the orthogonality condition we focus on is model dependent. Moreover, the structure of each model implies a specific ranking as to how the estimates of θ should relate to each other across models. Ultimately, however, how different margins of trade affect estimates of the welfare gains from trade is a quantitative question, which we turn to next.

5 Estimation Procedure

This section describes how we estimate the parameter that governs the trade elasticity in each model given a data set that features micro-level prices and bilateral trade flows across countries. The basic idea is captured in Proposition 3: we use moment conditions that compare statistics from a sample of micro-level prices with their expected values from each model, where the latter are functions of θ and depend on assumptions about the model that generated the data. Because the expected values do not have closed form expressions, we use simulation methods to approximate the expected value and we discuss how we are able to simulate prices from the different models. In Appendix B, we provide monte-carlo evidence that our estimation procedure works well.

5.1 Estimation

We will focus on three sets of results (i) an exactly identified case, (ii) an overidentified case with an identity weighting matrix, and (iii) an overidentified case with an optimal weighting matrix. Below we discuss each of these cases in turn.

Exactly Identified Case. In the exactly identified case we focus on properties of the logarithm of the maximal price gap adjusted by average log prices in the two locations.

Following the empirical exercise in Section 2, we define the object d_{ni} in each model as

$$d_{ni} = \log \hat{\tau}_{ni} + \log \hat{P}_i - \log \hat{P}_n, \quad (12)$$

$$\text{where } \log \hat{\tau}_{ni} = \max_{\ell \in L} \{\log p_n(\ell) - \log p_i(\ell)\},$$

$$\log \hat{P}_i = \frac{1}{L} \sum_{\ell=1}^L \log(p_i(\ell)),$$

and L is the number of micro-level prices in the sample. Like the statistics discussed in Section 4, we focus initially on the maximal price gap.

One difference from the previous section is that equation (12) adjusts the maximal price gap by average price differences in the two locations. There are empirical and theoretical reasons for this adjustment. As stated in Section 2, empirically, adjusting for price levels in each country helps us control for country specific factors that may be present in the price data but not in our model.⁸ Theoretically, if one wants to use the maximal price gap as a proxy for trade frictions and connect this value with trade flows, then expression (7) suggests adjusting for the difference in Φ s across countries as well. It turns out that the difference in the means of logged prices equals the difference in the logs of the parameters Φ defined in expression (4).⁹

Given the moments $d_{ni,t}^d$ from data, we define the following function

$$h(\theta, d_{ni,t}^d, \mathcal{M}) = d_{ni,t}^d - \frac{1}{S} \sum_{s=1}^S d_{ni}(\theta, u_s, \mathcal{M}), \quad (13)$$

which compares the observed value $d_{ni,t}^d$ in year t to its expected value in each model. Because we do not have a closed form expression for the expected value of d_{ni} in the model, $\mathbb{E}(d_{ni}; \{\theta, \mathcal{M}\})$, we approximate this value via simulation. S denotes the number of simulations and u_s is the simulation error. The next section provides the exact details behind the simulation, but the basic idea is the following: given a model $\mathcal{M}$ and a value of θ , we

⁸For example, suppose there are country-specific factors such as sales taxes, which are multiples of the goods' prices. Then, the maximum log price difference between countries n and i will reflect the log differences in the country-specific sales taxes. The tax wedge washes out once we add the difference in the means of logged prices between countries i and n .

⁹In Simonovska and Waugh (2014) we formally prove the result for the EK model. A similar result can be obtained for the Armington and BEJK models where price levels are also proportional to the price parameters as demonstrated in Proposition 1 above.

simulate prices, construct s synthetic data sets, and then construct simulated values for $d_{ni}(\theta, u_s, \mathcal{M})$. The expected value of d_{ni} is the average across simulations S .

In the exactly identified case, our estimation is based on the following orthogonality restriction

$$\mathbb{E}(h(\theta_o, d_{ni,t}^d, \mathcal{M})) = 0. \quad (14)$$

That is, at the true value θ_o , in expectation the difference between the observed $d_{ni,t}^d$ and its expected value should be equal to zero. The sample average of $h(\theta, d_{ni}, \mathcal{M})$ is

$$\bar{h}(\theta, \mathcal{M}) = \left(\frac{1}{N^2 - N} \right) \sum_n \sum_i h(\theta, d_{ni}, \mathcal{M}) \quad (15)$$

where $N^2 - N$ is the number of d_{ni} s we can construct. Our estimate of θ is the value that minimizes the quadratic form of $\bar{h}(\theta, \mathcal{M})$. Or mathematically:

$$\hat{\theta}(\mathcal{M}) = \arg \min_{\theta} \bar{h}(\theta, \mathcal{M})' \bar{h}(\theta, \mathcal{M}). \quad (16)$$

Overidentified Case. The overidentified case focuses on two additional moments: (i) the price gap in the 85th percentile adjusted by average prices and (ii) the covariance of d_{ni} with the logarithm of bilateral distance.

The focus on the price gap in the 85th percentile is a way to incorporate more information about the underlying distribution of price gaps and, hence, information about the underlying model. For example, as discussed in Section 4, the two country Armington model places a restriction on the data that the max should be the same as the price gap in the 85th percentile. The price gap in the 85th percentile is not the only statistic that has this property. We explored other percentiles (e.g. 90 or 75) and found virtually no effect on our results.

The covariance between the d_{ni} statistic and the logarithm of distance was chosen for two reasons. First, as we show in Section 2, there is a strong correlation between maximal price gaps and distance. Given the strong role that distance plays in explaining trade flows, we feel this is a natural statistic to target. Second, this also helps guard against estimating the elasticity off of measurement error in the data by focusing on the systematic relationship between price gaps and distance.

Given these moments, the function $\mathbf{h}(\theta, \mathbf{D}_{ni,t}^d, \mathcal{M})$ is now a 3×1 vector. The first element of $\mathbf{D}_{ni,t}^d$ is $d_{ni,t}^d$ as defined in expression (12) above, the second is the price gap in the 85th percentile, $d_{ni2,t}^d$, and the third is the covariance between $d_{ni,t}^d$ and the logarithm of bilateral

distance. Our orthogonality restriction is

$$\mathbb{E}(\mathbf{h}(\theta_o, \mathbf{D}_{ni,t}^d, \mathcal{M})) = 0. \quad (17)$$

The sample average of (each element of) $\mathbf{h}(\theta, \mathbf{D}_{ni,t}^d, \mathcal{M})$ is

$$\bar{\mathbf{h}}(\theta, \mathcal{M}) = \left(\frac{1}{N^2 - N} \right) \sum_n \sum_i \mathbf{h}(\theta, \mathbf{D}_{ni,t}^d, \mathcal{M}), \quad (18)$$

and we chose θ to minimize the quadratic form

$$\hat{\theta}(\mathcal{M}) = \arg \min_{\theta} \bar{\mathbf{h}}(\theta, \mathcal{M})' \mathbf{W}^{-1} \bar{\mathbf{h}}(\theta, \mathcal{M}), \quad (19)$$

where $\mathbf{W}$ is a positive definite weighting matrix.

We use the optimal weighting matrix suggested by Gouriéroux and Monfort (1996) and described in Adda and Cooper (2003). Because the optimal weighting matrix depends on our estimate of θ , we use a continuous-updating estimator of W which continually updates the weighting matrix within the minimization routine (see e.g. the discussion of this estimator in Hansen et al. (1996)).

We compute standard errors using a parametric bootstrap technique (see e.g. Davison and Hinkley (1997)). We add error terms to the trade data and re-estimate the country-specific parameters from the simulated data necessary to simulate prices. Given these parameters, we simulate a sample of prices given our estimate of θ and the assumed underlying model, and then implement our estimation routine on the simulated data. Thus, this procedure takes into account uncertainty in the technology parameters and scaled trade costs estimated from trade data and sampling variability in the micro-level prices. We repeat this procedure 100 times to construct 90-10 percentile confidence intervals.

In the instances where we use an optimal weighting matrix, we also perform tests of overidentifying restrictions. Specifically, we construct the test statistic

$$(N^2 - N) \bar{\mathbf{h}}(\hat{\theta}, \mathcal{M})' \mathbf{W}^{-1} \bar{\mathbf{h}}(\hat{\theta}, \mathcal{M}) \quad (20)$$

which is the standard “J-statistic” used to test the null-hypothesis that the data generating process is correct. Asymptotically, this test statistic is distributed chi-squared with a certain number of degrees of freedom, though in small samples this may not provide an accurate approximation (see e.g. Hansen et al. (1996)). To avoid these difficulties, we use a parametric bootstrap to construct the finite sample distribution of the test statistic in (20) under the null hypothesis that the data generating process is correct (see Davison and Hinkley (1997)).

BEJK Details. One detail associated with the BEJK model is that the value of ρ , the CES parameter, matters unlike in the EK model. The issue is that ρ determines the monopoly pricing rule which shapes the distribution of price gaps. We calibrate this parameter to the value of 2.5 (which is consistent with evidence regarding this parameter in Broda and Weinstein (2006)) and we estimate only θ . We then show robustness results under different values for ρ .

Relation to Previous Work. The statistic d_{ni} that we use to estimate θ has a history in the literature, so a couple of comments are in order. This statistic corresponds to the (demeaned) maximal price difference across goods for a pair of countries. EK used it to directly estimate θ : if the maximal price difference accurately reveals the trade friction, then averaging across normalized trade shares relative to the average d_{ni} yields an estimate of θ , as can be seen from equation (6). Using this approach, they arrived at a value of 8.28. As we demonstrate in Simonovska and Waugh (2014), however, the maximal price gap generally underestimates the trade friction and hence overestimates θ . To address this, we developed a simulation methodology that estimates θ by targeting the biased EK estimates directly, using price data for 123 countries in 2004. Following this method, we arrived at a value of four.

In this paper, we replace the approach of targeting the biased EK estimates and instead directly target moments related to the d_{ni} statistics themselves. This approach is more transparent, and we further enrich it with a moment that captures the covariance between the maximum price gap and the log of bilateral distance. The motivation to examine these moments comes from our empirical findings in Section 2, where price gap moments vary systematically with standard gravity variables that explain observed trade patterns. We then use these moments to infer trade elasticity parameters across different models and to test the models' fit to the data. In the exactly identified case, our new approach gives nearly the same answer for the EK model as the approach in Simonovska and Waugh (2014), which is reassuring.

Why these Moments. The preceding theoretical and empirical results provided a tactical rationale for why we chose the particular moments we did. We have carefully chosen the moments, explored a wide variety of alternatives (such as different percentiles, order statistics from different-sized samples via recurrence relationships, etc.) and found similar results.

There are, however, deeper, strategic issues regarding the choice of moments that are worth pointing out. One may wonder why we do not use other micro moments such as the firm size/sales distribution. The reasoning is simple: Moments about the firm/sales size distribution are indeterminate in the Armington and EK models. In contrast, all models make concrete and distinct predictions about price variation; hence, moments from the price

distribution are informative.

One may also wonder why we do not find and then use information regarding the existence of traded versus non-traded goods for the identification of trade elasticities. If we could find a good where the arbitrage relationship bound, under the assumption of perfect competition, this good's price difference could be used to estimate a trade friction, and consequently, a trade elasticity. While this information may help, it does not help one answer the central question: Are parameters that govern the trade elasticities (and therefore welfare gains) different across models that fit the same moments in the data? For example, if the true data generating process is BEJK, then prices reflect variable mark-ups, so cross-country price differences in *traded* goods are not necessarily equal to trade costs as discussed in Section 4. Furthermore, focusing on traded goods does not inform us about the underlying data generating process. As the arguments in Section 4 suggest and Figure 1 makes clear, the data features that distinguish different models are price gaps *away* from the trade costs. Moreover, the economic mechanisms (margins of trade) behind each of these models shape the pattern of price gaps *away* from the trade costs. This is the exact rationale for focusing on moments from the price gap distribution that are away from the trade cost, such as the 85th percentile of the price gap.

Sources of error. Our interpretation of the error between model-predicted and observed price gaps is that it reflects sampling variability which is a function of the underlying trade model. Given the underlying trade model, prices are realizations of random variables sampled from the equilibrium distribution of prices, and this sampling creates the main source of error between the model moment and the data analog.

The other source of error, which we discuss below, comes from the fact that the gravity regression does not perfectly match the trade data. Thus, the parameters which we estimate from the gravity regression and which are used in the simulation approach, are measured with error as well. When constructing confidence intervals, this source of error is taken into account.

There are other possible sources of error that we do not incorporate into the estimation. One is measurement error, which we addressed in Simonovska and Waugh (2014). There, we argued that if measurement errors were large, we should see little systematic relationship between price gaps and gravity variables such as distance. And that is not what we find. Moreover, we are building directly on this point by incorporating the covariance of price gaps with log distance as a third moment in our estimation approach. Heterogeneous trade costs, and in particular, trade costs correlated with productivity are another possibility. The main reason we do not pursue this is to remain purely within the ACR class of models. We are not aware of modeling approaches that accommodate this within that class.

5.2 Simulation Approach

A key step in the estimation procedure is the approximation of $\mathbb{E}(d_{ni}; \{\theta, \mathcal{M}\})$ via simulation. Below we describe in detail how we simulate prices from each model and in turn how we approximate $\mathbb{E}(d_{ni}; \{\theta, \mathcal{M}\})$.

There are three steps. The distribution of prices depends on θ and the country-specific technology parameters and bilateral trade costs. We use the common gravity equation implied by all models to estimate the technology parameters and trade costs (up to a scalar θ) from trade flow data. This ensures that all models have the same aggregate predictions for trade flows. Second, we show how the estimates from the gravity equation are sufficient to simulate micro-level prices given a specified model. Third, we specify a protocol as to how the data is sampled. These three steps allow us to create synthetic data sets from which we can then approximate $\mathbb{E}(d_{ni}; \{\theta, \mathcal{M}\})$.

Step 1.—We estimate the parameters for the country-specific productivity distributions and trade costs from bilateral trade-flow data. We follow closely the methodologies proposed by Eaton and Kortum (2002) and Waugh (2010). First, we derive the gravity equation from expression (3) by dividing the bilateral trade share by the importing country's home trade share,

$$\log\left(\frac{X_{ni}/X_n}{X_{nn}/X_n}\right) = S_i - S_n - \theta \log \tau_{ni} + \nu_{ni}, \quad (21)$$

where S_i is defined as $\log[T_i w_i^{-\theta}]$. S_i 's are recovered as the coefficients on country-specific dummy variables given the restrictions on how trade costs can covary across countries. We assume that ν_{ni} reflects other factors and is orthogonal to the regressors and normally distributed with mean zero and standard deviation σ_ν . ν_{ni} is another source of error that we account for when we perform statistical inference.

Trade costs take the following functional form:

$$\log \tau_{ni} = d_k + b_{ni} + im_n. \quad (22)$$

Here, trade costs are a logarithmic function of distance, where d_k with $k = 1, 2, \dots, 6$ is the effect of distance between country i and n lying in the k -th distance interval.¹⁰ b_{ni} is the effect of a shared border in which $b_{ni} = 1$ if country i and n share a border and zero otherwise.

The term im_n is an importer fixed effect which allows the trade-cost level to vary depending upon the importer and this specification follows EK. As we showed in Simonovska

¹⁰Intervals are in miles: $[0, 375)$; $[375, 750)$; $[750, 1500)$; $[1500, 3000)$; $[3000, 6000)$; and $[6000, \text{maximum}]$.

and Waugh (2014), using exporter fixed effects (as in Waugh (2010)) or aggregate price data to exactly identify trade costs did not change our estimates by economically meaningful amounts. Finally, our results are robust to incorporating bilateral colonial, language, and legal origin ties as well as countries' geographical attributes. We use least squares to estimate equations (21) and (22).

Step 2.—The parameter estimates obtained from the first-stage gravity regression are sufficient to simulate micro-level prices in each model up to a constant, θ .

Step 2a.—The simulation of the Armington model is simple as there is no micro-level heterogeneity in this framework. $\exp(S_i^{-\theta})$ represents the common marginal cost of production across all producers in country i . We assume that a product j in the model corresponds to a basic heading j in the data.

Step 2b.—To simulate micro-level prices in the EK model we follow Simonovska and Waugh (2014). We assume that a product j in the model corresponds to a basic heading j in the data. Notice that for any good j , the model implies that $p_{ni}(j) = \tau_{ni}w_i/z_i(j)$. Thus, rather than simulating productivities, it is sufficient to simulate the inverse of marginal costs of production $u_i(j) = z_i(j)/w_i$. In Simonovska and Waugh (2014), we show that u_i is distributed according to:

$$M_i(u_i) = \exp(-\tilde{S}_i u_i^{-\theta}), \text{ with } \tilde{S}_i = \exp(S_i) = T_i w_i^{-\theta}. \quad (23)$$

Thus, having obtained estimates of S_i from the gravity regression, we can simulate the inverse of marginal costs. To construct prices at which goods are traded, we take the inverse marginal costs that are drawn from the country-specific distributions above and are assigned to each good. Then, for each importing country and each good, the lowest-cost supplier across countries is found and realized prices are recorded.

Step 2c.—In BEJK, we assume that a product j in the model corresponds to a basic category j in the data. To see that we can simulate micro-level prices in the BEJK model as a function of θ only, we draw on an argument in BEJK. To simulate their model, BEJK reformulate the model in terms of efficiency. In particular, given two productivity draws for good j in country i , $z_{1i}(j)$ and $z_{2i}(j)$, they define the following objects:

$$\begin{aligned} u_{1i}(j) &= T_i z_{1i}(j)^{-\theta} \\ u_{2i}(j) &= T_i z_{2i}(j)^{-\theta} \end{aligned}$$

BEJK demonstrate that these objects are distributed according to

$$\begin{aligned} Pr[u_{1i} \leq u_1] &= 1 - \exp(-u_1) \\ Pr[u_{2i} \leq u_2 | u_{1i} = u_1] &= 1 - \exp(-u_2 + u_1) \end{aligned}$$

To simulate trade flows and prices from the model, we define the following variables

$$\begin{aligned} v_{1i}(j) &= \left(\frac{u_{1i}(j)}{T_i w_i^{-\theta}} \right) \\ v_{2i}(j) &= \left(\frac{u_{2i}(j)}{T_i w_i^{-\theta}} \right) \end{aligned}$$

Applying the pdf transformation rule, it is easy to demonstrate that $v_{1i}(j)$ is distributed according to

$$Pr[v_{1i} \leq v_1] = 1 - \exp(-\tilde{S}_i v_1). \quad (24)$$

Similarly,

$$Pr[v_{2i} \leq v_2 | v_{1i} = v_1] = 1 - \exp(-\tilde{S}_i v_2 + \tilde{S}_i v_1), \quad (25)$$

where $\tilde{S}_i$ is defined above.

Thus, to simulate the model we draw minimum unit costs from (24), and conditional on these draws, we draw the second lowest unit costs from (25). Hence, having obtained the estimates of S_i from the first-stage gravity regression, we can simulate the inverse of marginal costs.

To construct prices at which goods are traded, we take the inverse marginal costs that are drawn from the country-specific distributions above and are assigned to each good. Then, for each importing country and each good, the two lowest-cost suppliers across countries are found and realized prices are recorded.

Step 3.—From the realized prices, a subset of goods common to all countries is defined and the subsample of prices is recorded—i.e., we are acting as if we were collecting prices for the international organization that collects the data. The three models give a natural common basket of goods to be priced across countries. In these models, agents in all countries consume all goods that lie within a fixed interval, $[0, 1]$. Thus, we consider this common list in the simulated models and we randomly sample the prices of its goods across countries.

These steps then provide us with an artificial data set of micro-level prices that mimic

Table 4: Baseline Results: Exactly Identified

Model	Year	θ	CI (p10–p90)
BEJK ($\rho = 2.5$)	2004	3.28	3.13 – 3.36
BEJK ($\rho = 2.5$)	2011	3.19	3.08 – 3.30
BEJK ($\rho = 2.5$)	2017	3.33	3.21 – 3.43
EK	2004	4.09	3.92 – 4.23
EK	2011	3.94	3.81 – 4.08
EK	2017	4.19	4.05 – 4.34
Armington	2004	5.19	4.96 – 5.57
Armington	2011	4.90	4.73 – 5.12
Armington	2017	5.17	4.94 – 5.41

Notes: Single moment condition: $\mathbb{E}[d_{ni,t}^d - d_{ni}(\theta, \mathcal{M})] = 0$. Confidence intervals are (p10–p90) from parametric bootstrap with 100 replications.

their analogs in the data. Given this artificial data set, we can then compute moments—as functions of θ —and compare them to the moments in the data.

In order to assess the validity of our approach, in Section B, we apply the estimation methodology on data simulated by each model under a known value for θ , and we find that we recover the value for each model under the exactly identified and overidentified strategies. Therefore, we proceed to estimate the elasticity parameters for each model using our dataset from Section 2 next.

6 Estimation Results

Table 4 presents the results for the exactly identified specification, which most closely corresponds with our theoretical results in Proposition 3. In this specification, each model perfectly matches the single moment that it targets in the data, $d_{ni,t}^d$, in every year.

Consistent with the theoretical results, the estimate of θ is systematically lower for the new trade models relative to the Armington model, and further the estimate for BEJK is lower than for EK. The difference in magnitudes is substantial. The EK estimate is about 20 percent lower than Armington, implying that the estimated welfare cost of autarky is 20 percent higher in the EK model. In comparison, our estimate of the trade elasticity in the BEJK model is about 20 percent lower relative to EK, implying the welfare gains are estimated to be 20 percent larger in BEJK relative to EK. Focusing on each row of the table, it is evident that the estimates are very stable across the different time samples. For a given model, the estimates are not statistically different from each other over time.

Table 5 presents the results for the overidentified case when the optimal weighting ma-

Table 5: Baseline Results: Over-Identified

Model	Year	θ	CI (p10–p90)	J -stat	J pctlile	Max gap (mod/data)	85% gap (mod/data)	cov(max, log(dist)) (mod/data)
BEJK ($\rho=2.5$)	2004	3.13	3.08 – 3.20	86.3	1.00	0.98 / 0.93	0.35 / 0.37	0.09 / 0.09
BEJK ($\rho=2.5$)	2011	3.13	3.07 – 3.21	192.5	1.00	1.00 / 0.98	0.34 / 0.34	0.09 / 0.25
BEJK ($\rho=2.5$)	2017	3.24	3.18 – 3.32	183.9	1.00	0.94 / 0.91	0.32 / 0.32	0.07 / 0.20
EK	2004	4.00	3.91 – 4.12	10.7	0.38	0.95 / 0.93	0.36 / 0.37	0.09 / 0.09
EK	2011	4.07	3.99 – 4.18	136.8	0.99	0.95 / 0.98	0.34 / 0.34	0.09 / 0.25
EK	2017	4.18	4.09 – 4.31	139.3	1.00	0.91 / 0.91	0.32 / 0.32	0.07 / 0.20
Armington	2004	3.76	3.56 – 3.92	1390.5	1.00	1.28 / 0.93	0.25 / 0.37	0.23 / 0.09
Armington	2011	3.49	3.34 – 3.60	983.9	1.00	1.38 / 0.98	0.25 / 0.34	0.23 / 0.25
Armington	2017	3.82	3.64 – 3.98	1411.7	1.00	1.23 / 0.91	0.21 / 0.32	0.20 / 0.20

Notes: Three moment conditions: $d_{ni,t}^d$ (max price gap), $d_{ni,2,t}^d$ (85th percentile price gap), $\text{cov}(d_{ni,t}^d, \log \text{distance})$. Optimal weighting matrix. J pctlile is the percentile of the J -statistic in the bootstrap distribution.

trix is used. Our estimate for the Armington model declines substantially relative to the previous table and even falls below the EK estimate. This may seem problematic relative to our results in Proposition 3, yet in fact it is highly symptomatic of the fact that the Armington model is misspecified. The last three columns of the table illustrate this point. Across all moments, the Armington model has problems: it overpredicts the mean order statistic by a large margin and it underpredicts the 85th percentile by a large margin. The covariance moment is lower in year 2004 compared to 2011 and 2017, and the Armington model matches it closely only in that year.

The difficulties of the Armington model are formalized in the P-value for the J-statistic which tests the null hypothesis that the Armington model is the data generating process. For the Armington model, the null hypothesis is rejected at a level of less than one percent and J-statistic is astronomically high. Given the starkness of the Armington model, this should be a reassuring result.

For the EK model, the results with the optimal weighting matrix are nearly identical to the results from the exactly identified specification. However, the model is also rejected by the data, with the exception of year 2004, where the model closely fits all three moments. The EK model exactly matches the 85th percentile (which is given the most weight) and it comes very close to matching the max. It fails to match the covariance term, with the exception of year 2004. The J-statistic formalizes this and we fail to reject the null hypothesis in 2004 that the EK model is the data generating process.

Across specifications, the results for the BEJK model are also similar. Although the BEJK model is the richest of the three models in terms of market structure and pricing patterns, there is evidence that it is misspecified as the null hypothesis that the BEJK model generated the data is rejected. The BEJK model struggles more to match the max and the covariance term than the EK model. The problem is due to variable markups. Variable markups naturally induce “noise” between price gaps (even if the good is traded in which case price gaps reflect the distance cost), which makes it harder for the BEJK model to replicate the strong covariance observed in the data between price gaps and distance.

Finally, the results for the BEJK model are sensitive to the assumed value for the parameter ρ . The estimated θ increases monotonically with ρ . Appendix C reports the results and we briefly discuss them here. Intuitively, as ρ rises, Bertrand markups shrink, reducing the model’s ability to compress price gaps, so a higher θ is needed. More importantly, at lower levels of ρ , the data no longer reject the BEJK model in year 2004. Therefore, it appears that the richest model comes closest to matching the price moments in the data when the trade elasticity falls below 3.

7 Endogenous Variety Models

The analysis thus far has focused on three canonical models of trade where the set of goods is fixed. Two other canonical models that feature an endogenous set of produced and consumed goods are the frameworks of Krugman (1980)—Krugman—and Melitz (2003)—Melitz. While these models can satisfy the same aggregate restrictions as the three models analyzed, we have kept them separate largely because we make additional assumptions about the structure of fixed costs and how we interpret the data when the set of goods varies across countries. Below, we outline versions of these two models and we repeat the quantitative exercises from above.

7.1 Krugman (1980)

Preferences are identical to the Armington model, but the measure of goods produced by any country i is endogenous,

$$W_n = \left[\sum_{i=1}^N \int_{J_i} x_{ni}(j)^{\frac{\rho_{\text{KRUG}}-1}{\rho_{\text{KRUG}}}} dj \right]^{\frac{\rho_{\text{KRUG}}}{\rho_{\text{KRUG}}-1}},$$

where $\rho_{\text{KRUG}} > 1$ and J_i denotes the set of goods produced by country i .

In particular, we assume that a producer in country i has monopoly power over a blueprint to produce a single good j . In order to produce the good, a firm in i uses the following production technology

$$y_i(j) = T_i^{1/\theta_{\text{KRUG}}} l_i(j) + e_i,$$

where $\theta_{\text{KRUG}} > 0$ is a scaling parameter and $l_i(j)$ denotes labor. A firm incurs a marginal cost of $w_i/T_i^{1/\theta_{\text{KRUG}}}$ as well as a fixed cost of entry $e_i > 0$, both in labor units. There is an unbounded pool of potential producers and the mass of firms is pinned down via a zero average profit condition.

We will consider two versions of the Krugman model that differ in the assumptions about the fixed cost of entry. The two specifications will allow us to evaluate the role of the home market effect in this model.

In the first version of the Krugman model, we assume that fixed entry costs are proportional to the mass of workers in country i , L_i . Under the proportionality assumption on fixed costs, the mass of entrants is identical across countries and proportional to $1/(1 + \theta)$.

Conditional on entry, all consumers buy all products from all sources. Moreover, due to the proportionality assumption on entry costs, each country contributes a measure of $1/N$ toward the unit measure of consumed products as in the Armington model. In contrast to Armington, the price $p_{ni}(j)$ for good j from i in destination n is given by the product of the marginal cost of production and delivery and the Dixit-Stiglitz constant mark-up $\bar{m}$. Since mark-ups are not country-specific, relative prices of identical goods across countries reflect relative marginal costs of production only. Therefore, this version of the Krugman model maps directly into the multi-country symmetric Armington model that we outlined in Section 3.

This version of the Krugman model, however, shuts down the market size effect. In order to evaluate the role that this channel plays in the estimates of the trade elasticity parameter in the Krugman model, we assume that the fixed entry cost $e_i = e \forall i$. In this case, the measure of entrants from each country is exactly proportional to the size of the market, L_i . Hence, each country i contributes exactly $L_i/\sum_k L_k$ fraction of products toward the unit measure of available products.

7.2 Melitz (2003)

Finally, we outline a variant of the Melitz (2003) model parameterized as in Chaney (2008). In the exposition, we follow closely Arkolakis et al. (2008).

The market structure and the preference relation, with parameter $\rho_{\text{MEL}} > 1$, are identical

to the Krugman (1980) model described above. There are two novel features. First, in order to sell its product to market n , a firm from country i incurs an additional market access cost, a_{ni} . Second, marginal costs of production are not identical across producers within a country. In particular, upon paying the entry cost e_i —which will take on the same two specifications as in the Krugman model—a firm from i draws a productivity realization for good j , $z_i(j)$, from the country-specific Pareto distribution with density $T_i z^{-\theta_{\text{MEL}}}$. As in the Frechet distribution, the parameter $\theta_{\text{MEL}} > 0$ in the Pareto distribution governs the variability of productivity among firms.

The two additional assumptions imply that there exist cost thresholds that limit the participation of certain firms to certain markets. In particular, if a firm obtains a cost draw above the threshold cost that characterizes the “most accessible” destination, it exits immediately without operating. Thus, because of free entry, in equilibrium expected profits of a firm must be equal to the fixed entry cost. In addition, more productive firms serve more markets and the “toughest” markets attract the most efficient producers from all sources.

As in the Krugman (1980) model, firms charge a constant mark-up over their marginal cost of production. However, unlike the Krugman (1980) model, the prices of goods offered by a given country differ according to the efficiency of the individual producer. Additionally, selection among exporters alters the composition of the bundle of consumed goods in every country.

We assume that the market access cost $a_{ni} = \xi X_n (\forall i)$, where $\xi > 0$ is a country-invariant parameter and $X_n = w_n L_n$ is country- n expenditure. This assumption features the special case where market access costs are expensed in destination wages (Ignatenko et al., 2025) and are proportional to market size, L_n . The assumption will effectively shut down the market size effect portion of the extensive margin adjustment across destinations. The market size effect of entry will continue to operate as long as the fixed entry cost is identical across countries as was the case in the second specification of the Krugman model above.

7.3 Trade Flows, Aggregate Prices, and Welfare

We begin by showing that the two models with an endogenous set of varieties produce identical aggregate outcomes to the models with an exogenous set of varieties under the assumption that $e_i \propto L_i (\forall i)$ and $a_{ni} = \xi X_n (\forall i, n)$. In particular, under the parametric assumption made above, all five models yield the same expressions for trade flows, price indices (up to a constant scalar), and welfare gains from trade. Proposition 4 summarizes the result. The proof can be found in Appendix A.

Proposition 4. *Assume $\theta_{\text{ARM}} = \rho_{\text{ARM}} - 1$, $\theta_{\text{KRUG}} = \rho_{\text{KRUG}} - 1$ and $\theta_{\text{MEL}} > \rho_{\text{MEL}} - 1$. Given the functional*

forms for productivity $F_{EK,i}(\cdot)$, $F_{BEJK,i}(\cdot)$, and $F_{MEL,i}(\cdot)$ for all $i = 1, \dots, N$, and assuming that (i) fixed entry costs are proportional to the mass of workers in the country of origin, $e_i \propto L_i$ for all $i = 1, \dots, N$; (ii) market access costs are proportional to market expenditure, $a_{ni} = \xi X_n$ for all $i, n = 1, \dots, N$;

- a. The share of expenditures that n spends on goods from i , X_{ni}/X_n , predicted by all five models is

$$\frac{X_{ni}}{X_n} = \frac{T_i(\tau_{ni}w_i)^{-\theta_M}}{\sum_{k=1}^N T_k(\tau_{nk}w_k)^{-\theta_M}}. \quad (26)$$

- b. The CES exact price index for destination n , P_n , predicted by all five models is

$$P_n \propto \Phi_n^{-\frac{1}{\theta}}, \quad \text{where} \quad \Phi_n = \sum_{k=1}^N T_k(\tau_{nk}w_k)^{-\theta_M}. \quad (27)$$

- c. The percentage compensation that a representative consumer in n requires to move between two trading equilibria predicted by all five models is

$$\frac{P'_n}{P_n} - 1 = 1 - \left(\frac{X'_{nn}/X'_n}{X_{nn}/X_n} \right)^{\frac{1}{\theta_M}}, \quad (28)$$

where $M = \text{ARM}, \text{EK}, \text{BEJK}, \text{KRUG}, \text{MEL}$.

The proof appears in Appendix A and entails deriving a solution for the Krugman and Melitz models under the stated parametric restrictions.

In contrast, if we assume that $e_i = e(\forall i)$ and $a_{ni} = \xi X_n$ for all $i, n = 1, \dots, N$, the two endogenous variety models yield identical expressions for trade flows and price indices, where the expression Φ_i is replaced by Ψ_i , which further reflects country-size differences. Nevertheless, the expressions characterizing the welfare gains from trade are unchanged. Proposition 5 summarizes the result. The proof can be found in Appendix A.

Proposition 5. Assume $\theta_{\text{KRUG}} = \rho_{\text{KRUG}} - 1$ and $\theta_{\text{MEL}} > \rho_{\text{MEL}} - 1$. Given the functional form for productivity $F_{\text{MEL},i}(\cdot)$ for all $i = 1, \dots, N$, and assuming that (i) fixed entry costs are country invariant, $e_i = e$ for all $i = 1, \dots, N$; (ii) market access costs are proportional to market expenditure, $a_{ni} = \xi X_n$ for all $i, n = 1, \dots, N$;

- a. The share of expenditures that n spends on goods from i , $\tilde{X}_{ni}/\tilde{X}_n$, predicted by the Krugman

and Melitz model is

$$\frac{\tilde{X}_{ni}}{\tilde{X}_n} = \frac{L_i T_i (\tau_{ni} w_i)^{-\theta_M}}{\sum_{k=1}^N L_k T_k (\tau_{nk} w_k)^{-\theta_M}}. \quad (29)$$

b. The CES exact price index for country n , $\tilde{P}_n$, predicted by the Krugman and Melitz model is

$$\tilde{P}_n \propto \Psi_n^{-\frac{1}{\theta_M}}, \quad \text{where} \quad \Psi_n = \sum_{k=1}^N L_k T_k (\tau_{nk} w_k)^{-\theta_M}. \quad (30)$$

c. The percentage compensation that a representative consumer in n requires to move between two trading equilibria predicted by all five models is

$$\frac{\tilde{P}'_n}{\tilde{P}_n} - 1 = 1 - \left(\frac{\tilde{X}'_{nn}/\tilde{X}'_n}{\tilde{X}_{nn}/\tilde{X}_n} \right)^{\frac{1}{\theta_M}}, \quad (31)$$

where $M = KRUG, MEL$.

7.4 Simulation of Prices for Endogenous Variety Models

Since the endogenous variety models yield an identical gravity equation of trade as the exogenous variety models, the first stage of the estimation is identical. What remains is to describe the micro price-level simulation.

Step 2d.—The simulation of both versions of the Krugman model is trivial because the model does not feature micro-level heterogeneity. In the version outlined in Proposition 4, $S_i \equiv T_i w_i^{-\theta}$ fully characterizes the common marginal cost of production across all producers in country i . In the version outlined in Proposition 5, the following change of definition occurs: $S_i \equiv L_i T_i w_i^{-\theta}$. Thus, the simulation is independent of the fixed entry cost assumption. We map a good j in the Krugman model to a basic heading j in the data, and we compare the relative price of that good across countries. Hence, good j produced by country i is distinct from good j' produced by country k . However, in the absence of heterogeneity, all goods from i sold to a destination n have the same price, which purely reflects the bilateral trade barrier, the constant mark up and the marginal cost from country i .

Step 2e.—The simulation of the Melitz (2003) model is more intricate and depends on the various parametric assumptions. In the Eaton and Kortum (2002) and Bernard et al. (2003) models—which feature heterogeneity—a good is indexed by $j \in [0, 1]$ or an integer when discretized on the computer. In the Melitz (2003) model, different countries are consuming and producing different goods. As in the Krugman model, in the version of Melitz

outlined in Proposition 4, $S_i \equiv T_i w_i^{-\theta}$ fully characterizes the common marginal cost of production across all producers in country i . Also, as in Krugman, good j produced by country i is distinct from good j' produced by country k . But, we need to keep track of each individual good and compare its price across destinations due to good-specific marginal costs.

To carry out the simulation, we discretize the continuum to a set of potentially produced goods. We proceed to determine the subset of goods produced domestically for each country. Then, we simulate inverse marginal costs for each good. Inverse marginal costs are determined following Eaton et al. (2011) who show how normalized inverse marginal costs can be sampled from a parameter free uniform distribution. This procedure is made possible by the parametric assumption that we impose on the market access cost, which effectively makes threshold costs destination specific (see derivations in proofs to Propositions 4 and 5 in Appendix A). Then, we determine the set of exported goods for all country-pairs and we compute their prices.

As in the exogenous variety models, we must specify how the subsample of prices is recorded. In the fixed variety models, because all goods are consumed everywhere, there is a natural common list from which we randomly sample the prices of its goods across countries. Several issues arise in the endogenous variety models, and we discuss them below.

Step 3a.—In the Krugman model where $e_i \propto L_i$ ($\forall i$), all produced goods appear in all countries, and each country contributes a fraction $1/N$ of them. Thus, like the exogenous variety models, there is a natural common basket of goods prices to be randomly sampled from across countries. In the case in which $e_i = e$ ($\forall i$), the same random sampling procedure of goods will result, in expectation, in different number of goods from different countries appearing on the list of common goods. The intuition is simple. Larger countries produce more goods, and the ratio L_i/L_n governs the relative measure of goods available from countries i and n . Since all goods are exported everywhere, this means that there will be L_i/L_n more goods from country i on the common list relative to country n . So the probability that a random draw will reflect a good from country i is higher if $L_i > L_n$.¹¹

Step 3b.—We simulate the version of the Melitz model corresponding to the assumptions in Proposition 4. This is our baseline scenario that allows for direct comparison with the fixed variety models. We do not simulate the second version of the Melitz model, since we evaluate the role of the market size effect within the context of the Krugman model.

In the Melitz model, we define the common set of goods whose prices will be sampled to be the set of goods produced by firms with low enough cost draws so that they serve all N

¹¹This gives us a natural rule to follow when we simulate goods' prices in the Krugman model with the home market effect. Specifically, if each country has $M_i = M$ goods in the simulation, we reshuffle the set of goods that each country contributes to the common list by weight $L_i/\sum_k L_k$. Then, larger countries contribute larger shares to the list of common goods.

Table 6: Baseline Results: Exactly Identified

Model	Year	θ	CI (p10–p90)
Melitz	2004	3.50	3.06 – 3.90
Melitz	2011	3.30	3.10 – 3.83
Melitz	2017	3.32	2.83 – 3.73
Krugman-HM	2004	3.68	3.28 – 4.04
Krugman-HM	2011	3.62	3.31 – 4.12
Krugman-HM	2017	3.65	3.25 – 4.06
Krugman	2004	5.19	4.96 – 5.57
Krugman	2011	4.90	4.73 – 5.12
Krugman	2017	5.17	4.94 – 5.41

Notes: Single moment condition: $\mathbb{E}[d_{ni,t}^d - d_{ni}(\theta, \mathcal{M})] = 0$. Confidence intervals are (p10–p90) from parametric bootstrap with 100 replications.

destinations (see Appendix A for cost threshold derivations). Let M_i represent the measure of goods from i that appear on the common list and let $M = \sum_m M_m$ be the common list. Then, it is straightforward to derive the following expression for the share of country i 's goods on the common list

$$\frac{M_i}{M} = \frac{\min_l \tilde{S}_i \tau_{li}^{-\theta} \left[\sum_k \tilde{S}_k \tau_{lk}^{-\theta} \right]^{-1}}{\sum_m \min_l \tilde{S}_m \tau_{lm}^{-\theta} \left[\sum_k \tilde{S}_k \tau_{lk}^{-\theta} \right]^{-1}}. \quad (32)$$

Notice that this share varies across countries. This variation is precisely the reason behind the different elasticity estimates that we obtain from the Melitz relative to the baseline Krugman model as it directly reflects the selection mechanism in the Melitz model.

In the baseline Krugman model, the maximal bilateral price difference is likely to equal the unobserved trade friction (exactly like in the Armington model). In the Melitz model, only goods sold by firms that enter all markets are able to make the sample of prices that data collectors would consider. Moreover, the firms that enter all markets are likely to be from source countries with low trade costs to export. This implies that, in the Melitz model, the average maximal price difference across country pairs is likely to be below the average trade friction across country pairs. Thus, one needs a lower trade elasticity in Melitz relative to the baseline Krugman to rationalize the same amount of trade observed in the data. The differences in trade elasticities then translate into differences in the measured welfare gains from trade in the two models.

Table 7: Baseline Results: Over-Identified

Model	Year	θ	(p10–p90)	J -stat	J pctl	Moment 1 (mod/data)	Moment 2 (mod/data)	Moment 3 (mod/data)
Melitz	2004	3.59	3.38 – 3.91	133.2	0.86	0.90 / 0.93	0.34 / 0.37	0.21 / 0.09
Melitz	2011	3.15	2.91 – 3.67	60.8	0.53	1.03 / 0.98	0.29 / 0.34	0.25 / 0.25
Melitz	2017	3.08	2.60 – 3.82	287.5	0.92	0.98 / 0.91	0.23 / 0.32	0.25 / 0.20
Krugman-HM	2004	3.72	3.55 – 3.93	122.1	0.89	0.92 / 0.93	0.34 / 0.37	0.21 / 0.09
Krugman-HM	2011	3.39	3.03 – 3.57	45.5	0.42	1.05 / 0.98	0.31 / 0.34	0.25 / 0.25
Krugman-HM	2017	3.51	3.23 – 4.02	45.4	0.36	0.95 / 0.91	0.28 / 0.32	0.22 / 0.20
Krugman	2004	3.76	3.56 – 3.92	1390.5	1.00	1.28 / 0.93	0.25 / 0.37	0.23 / 0.09
Krugman	2011	3.49	3.34 – 3.60	983.9	1.00	1.38 / 0.98	0.25 / 0.34	0.23 / 0.25
Krugman	2017	3.82	3.64 – 3.98	1411.7	1.00	1.23 / 0.91	0.21 / 0.32	0.20 / 0.20

Notes: Three moment conditions: d_{ni} (max price gap), d_{ni2} (85th percentile price gap), $\text{cov}(d_{ni}, \log \text{distance})$. Optimal weighting matrix. J pctl is the percentile of the J -statistic in the bootstrap distribution.

7.5 Estimation Results

In Table 6, we report the results for the Krugman and Melitz models as specified in Proposition 4. In this case, the parametric assumptions shut down the market size effect of entry, although the selection effect operates in the Melitz model.

The Krugman model yields estimates identical to the symmetric Armington model, as is evident from the results in Table 4. Meanwhile, the Melitz model yields an estimate of θ that is consistently below the estimate for the Krugman model. Looking at the mean estimate across all three years, the Melitz model yields a parameter estimate that is at least 30% lower, suggesting that the measured welfare effects are much higher in this model.

We evaluate the role of the market size effect in the Krugman model, this is labeled Krugman-HM in Table 6 and Table 7. The result differs materially from the specification that shuts down the market size effect. In this richer specification, the elasticity estimate is only marginally smaller, but we can no longer reject the null hypothesis that the model is the data generating process. This finding suggests that the market size effect is a very important feature of the Krugman model.

Table 7 reports the results from the overidentified estimation with the optimal weighting matrix. As discussed earlier the Krugman/Armington model is rejected by the data given the J -stat presented in the table. The Melitz model yields moderately higher estimates than in the exactly identified case. More importantly, we can not reject the null hypothesis that the data are generated by the Melitz model in any of the three years.

Relation to Melitz and Redding (2015). Our estimation yields a lower trade elasticity estimate for the Melitz model than for the Krugman model, which translates into higher welfare gains from trade in the former. Melitz and Redding (2015) also argue that the Melitz model generates higher welfare gains from trade, but they place different constraints on the comparison. They hold common structural parameters fixed and then make a revealed preference argument: because the heterogeneous firm model has an additional margin of adjustment, it necessarily delivers higher welfare. Our approach does not fix the same parameters across models. Instead, we ask each model to match the same features of the data—both price gaps and bilateral trade shares. The reason welfare gains differ across models in our framework is that matching these moments requires different values of θ , and it is this difference in estimated trade elasticities that drives the welfare ranking.

8 Conclusion

We show that the welfare gains from trade are larger in models that feature more micro-level margins of adjustment, once these models are disciplined by data on both trade flows and micro-level prices. The key theoretical insight is that, although a well-known class of trade models all imply the same mapping from the trade elasticity to the welfare cost of autarky, they imply different mappings from the trade elasticity to the distribution of cross-country price gaps. Choosing the parameter that governs the trade elasticity to match the same price data therefore yields systematically different trade elasticities and welfare implications across models.

Quantitatively, among the fixed-variety models, the parameter governing the trade elasticity in BEJK is approximately 20 percent lower than in EK, and EK is 20 percent lower than Armington, implying a welfare cost of autarky in BEJK that is approximately 55 percent higher than in the intensive-margin-only benchmark. Among the endogenous variety models, the Melitz estimate is approximately 34 percent below the symmetric Krugman, but very similar to the estimate of the Krugman model that features the home market effect. These estimates are robust across three time periods of analysis. The following table summarizes these results. We hope it serves as a useful guide to the quantitative international trade literature.

Two additional results are worth highlighting. First, the BEJK estimates are sensitive to the CES elasticity of substitution, which governs the magnitude of variable markups. This sensitivity underscores that the mapping between the trade elasticity parameter and price data depends on the full model structure, not just on which margins are active. Second, an alternative specification of the Krugman model in which country size affects the gravity

Table 8: Summary Across Models (Exactly Identified, Baseline ρ)

Model	2004	2011	2017	Avg
BEJK ($\rho = 2.5$)	3.28	3.19	3.33	3.27
Melitz	3.50	3.30	3.32	3.37
Krugman (HM)	3.68	3.62	3.65	3.65
EK	4.09	3.94	4.19	4.07
Armington	5.19	4.90	5.17	5.09

equation yields substantially lower estimates of the trade elasticity parameter, as the market-size channel absorbs trade variation that would otherwise require a higher elasticity. This result illustrates more broadly that the welfare implications of trade depend on assumptions about market structure and fixed costs, even within models that share the same aggregate sufficient statistics.

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A Theory

The proof to Proposition 1 follows.

Proof: a. and b. We derive the trade shares and price index for the version of the Armington model that we consider in the present paper.

Maximizing U_n subject to a standard budget constraint yields the following CES demand function

$$x_{ni}(j) = \left(\frac{p_{ni}(j)}{P_n} \right)^{-\rho} \frac{w_n L_n}{P_n}, \quad (33)$$

$$P_n = \left[\sum_{i=1}^N \int_0^{1/N} p_{ni}(j)^{1-\rho} dj \right]^{\frac{1}{1-\rho}}. \quad (34)$$

Perfect competition ensures that pricing is according to marginal cost of production and delivery,

$$p_{ni}(j) = \frac{\tau_{ni} w_i}{T_i^{1/\theta}}. \quad (35)$$

Substituting (35) into (34), integrating, and letting $\theta = \rho - 1$ yields $P_n \propto \Phi_n^{-1/\theta}$, with proportionality constant $1/N$.

Finally, using (33) and (35), and integrating, the share of expenditure that n spends on goods from i is given by

$$\frac{X_{ni}}{\sum_{k=1}^N X_{nk}} = \frac{\left(\tau_{ni} w_i T_i^{-1/\theta} \right)^{1-\rho}}{\sum_{k=1}^N \left(\tau_{nk} w_k T_k^{-1/\theta} \right)^{1-\rho}} \quad (36)$$

Letting $\theta = \rho - 1$ yields the expression in the text. ■

The proof to Proposition 4 follows.

Proof: We only need to show the results for the endogenous variety models.

a. and b. We derive the trade shares and price indices for the versions of the Krugman (1980) and Melitz (2003) models that we consider in the present paper.

For the Krugman (1980) model, maximizing W_n subject to a standard budget constraint yields the following CES demand function

$$x_{ni}(j) = \left(\frac{p_{ni}(j)}{P_n} \right)^{-\rho} \frac{w_n L_n}{P_n}, \quad (37)$$

$$P_n = \left[\sum_{i=1}^N \int_{J_i} p_{ni}(j)^{1-\rho} dj \right]^{\frac{1}{1-\rho}}. \quad (38)$$

Moreover, maximizing firm profits and accounting for the demand in (37) yields

$$p_{ni}(j) = \frac{\bar{m} \tau_{ni} w_i}{T_i^{1/\theta}}. \quad (39)$$

Free entry implies that the expected profit of a firm is zero, after paying the fixed entry cost. Using the firm's optimal pricing rule and quantity demanded, which must equal to the quantity produced, yields

$$\sum_{n=1}^N \frac{(p_{ni}(j))^{1-\rho}}{P_n^{1-\rho}} \frac{w_n L_n}{\rho} - w_i e_i = 0 \quad (40)$$

Let M_i denote the measure of entrants in country i , which corresponds to the measure of set J_i . In order to compute the measure of entrants, combine the free entry condition from (40) with the labor market clearing condition for country i ,

$$M_i \sum_{n=1}^N \frac{(p_{ni}(j))^{1-\rho}}{P_n^{1-\rho}} w_n L_n + e_i = L_i, \quad (41)$$

to obtain $M_i = L_i / (\rho e_i)$. Using the definition $\theta = \rho - 1$ and assuming that $L_i \propto e_i$ yields $M_i \propto 1/(1 + \theta)$. Relying on this result, use the equilibrium measure of entrants M_i and the optimal pricing rule (39) into (38) to obtain that $P_n \propto \Phi_n^{-1/\theta}$.

Finally, the share of expenditure that n spends on goods from i is given by

$$\frac{X_{ni}}{\sum_{k=1}^N X_{nk}} = \frac{M_i \frac{(p_{ni}(j))^{1-\rho}}{P_n^{1-\rho}} \frac{w_n L_n}{\rho}}{\sum_{k=1}^N M_k \frac{(p_{nk}(j))^{1-\rho}}{P_n^{1-\rho}} \frac{w_n L_n}{\rho}} \quad (42)$$

Using the definition $\theta = \rho - 1$, the equilibrium measure of entrants M_i under the proportionality assumption, the optimal pricing rule (39) and the exact price index (38) in (42) yields the expression in the text.

For the Melitz (2003) model, following the same steps as for the Krugman (1980) model obtains the demand function in (37) and the price index in (38). Let the marginal cost of a firm with efficiency draw $z_i(j)$ of producing good j in i and delivering it to n be

$$c_{ni}(j) = \frac{w_i \tau_{ni}}{z_i(j)}.$$

We will express all equilibrium objects in terms of cost draws in the remainder of the Appendix. As in the Krugman (1980) model, the optimal pricing rule is given by

$$p_{ni}(j) = \bar{m} c_{ni}(j) \quad (43)$$

Using the firm's optimal pricing rule and quantity demanded, which must equal to the quantity produced, yields the firm's profit function

$$\pi_{ni}(c_{ni}(j)) = w_n L_n \left(\frac{\rho}{\rho - 1} \right)^{-\rho} \frac{1}{\rho - 1} \left(\frac{c_{ni}(j)}{P_n} \right)^{1-\rho} - \xi w_n L_n$$

The cost cutoff for a firm from i to sell good j to n , $\bar{c}_{ni}(j)$, satisfies $\pi_{ni}(\bar{c}_{ni}(j)) = 0$ and is given by

$$\bar{c}_{ni}(j) = \left[\xi (\rho - 1) P_n^{1-\rho} \left(\frac{\rho}{\rho - 1} \right)^\rho \right]^{\frac{1}{1-\rho}}$$

Since $\bar{c}_{ni}(j) = \bar{c}_{nn}(j) \forall i$, denote cost cutoffs by the destination to which they apply, $\bar{c}_n(j)$.

To determine the measure of entrants, repeat the procedure applied to the Krugman (1980) model to obtain $M_i = L_i / e_i (\rho - 1) / (\rho \theta)$. Only those firms with cost draws below $\bar{c}_i$ serve the domestic market and the remainder immediately exit. Using the Pareto distribution, the measure of domestic firms that sell j in country i is $M_{ii}(j) = M_i T_i w_i^{-\theta} (\bar{c}_i(j))^\theta$. Similarly, the measure of firms from i that sell j to n is $M_{ni}(j) = M_i T_i (w_i \tau_{ni})^{-\theta} (\bar{c}_n(j))^\theta$. Assuming $a_n = \xi w_n L_n$ makes the cutoff cost $\bar{c}_n(i)$ proportional to P_n .

Furthermore, assuming that $L_i \propto e_i$ yields $M_i \propto (\rho - 1) / (\rho \theta)$. Relying on this result, use the equilibrium measure of firms that serve destination n from any source i , M_{ni} , the optimal pricing rule (43), and the Pareto distribution into (38) to obtain that $P_n \propto \Phi_n^{-1/\theta}$.

Finally, to derive the trade share, use the optimal pricing rule (43), quantity demanded, which must equal to the quantity produced, the equilibrium measure of firms that serve each destination, and the Pareto distribution.

c. We refer the reader to Arkolakis et al. (2012) for derivations concerning welfare in all the models. ■

The proof to Proposition 5 follows.

Proof: The structure of each model is identical to Proposition 4; the only change is that $e_i = e$ for all i , rather than $e_i \propto L_i$.

a. and b. For the Krugman (1980) model, the CES demand function, price index, and optimal pricing rule are unchanged from (37)–(39). The free entry condition combined with the labor market clearing condition for country i yields $M_i = L_i/(\rho e)$. Since e is common across countries, $M_i \propto L_i$.

Substituting the optimal pricing rule (39) and $M_i \propto L_i$ into the price index (38) and using the definition $\theta = \rho - 1$ yields

$$\tilde{P}_n^{1-\rho} \propto \sum_{i=1}^N M_i \left(\bar{m} \tau_{ni} w_i T_i^{-1/\theta} \right)^{1-\rho} \propto \sum_{i=1}^N L_i T_i (\tau_{ni} w_i)^{-\theta} = \Psi_n.$$

Hence $\tilde{P}_n \propto \Psi_n^{-1/\theta}$.

The share of expenditure that n spends on goods from i is given by

$$\frac{\tilde{X}_{ni}}{\sum_{k=1}^N \tilde{X}_{nk}} = \frac{M_i \frac{(p_{ni}(j))^{1-\rho}}{\tilde{P}_n^{1-\rho}} \frac{w_n L_n}{\rho}}{\sum_{k=1}^N M_k \frac{(p_{nk}(j))^{1-\rho}}{\tilde{P}_n^{1-\rho}} \frac{w_n L_n}{\rho}} = \frac{M_i (p_{ni}(j))^{1-\rho}}{\sum_{k=1}^N M_k (p_{nk}(j))^{1-\rho}}.$$

Using $M_i \propto L_i$, the optimal pricing rule (39), and the definition $\theta = \rho - 1$ yields

$$\frac{\tilde{X}_{ni}}{\tilde{X}_n} = \frac{L_i T_i (\tau_{ni} w_i)^{-\theta}}{\sum_{k=1}^N L_k T_k (\tau_{nk} w_k)^{-\theta}} = \frac{L_i T_i (\tau_{ni} w_i)^{-\theta}}{\Psi_n}.$$

For the Melitz (2003) model, the demand function, price index, optimal pricing rule, and profit function are unchanged. The cost cutoff for a firm from i to sell good j to n satisfies $\pi_{ni}(\bar{c}_n(j)) = 0$ and is given by

$$\bar{c}_n(j) = \left[\xi(\rho - 1) \left(\frac{\rho}{\rho - 1} \right)^\rho \tilde{P}_n^{1-\rho} \right]^{\frac{1}{1-\rho}}.$$

Since $a_{ni} = \xi X_n$ for all i , $\bar{c}_n(j)$ depends only on the destination and is proportional to $\tilde{P}_n$.

The measure of entrants is $M_i = L_i/e \cdot (\rho - 1)/(\rho\theta)$. Since e is common across countries, $M_i \propto L_i$. The measure of firms from i that sell j to n is $\tilde{M}_{ni}(j) = M_i T_i (w_i \tau_{ni})^{-\theta} (\bar{c}_n(j))^\theta \propto L_i T_i (w_i \tau_{ni})^{-\theta} (\bar{c}_n(j))^\theta$.

Using $\tilde{M}_{ni}$, the optimal pricing rule (43), and the Pareto distribution in the price index (38) yields

$$\tilde{P}_n^{1-\rho} \propto \sum_{i=1}^N \tilde{M}_{ni}(j) \int_0^{\tilde{c}_n} (\tilde{m} c_{ni})^{1-\rho} \theta c_{ni}^{\theta-1} (\tilde{c}_n)^{-\theta} d c_{ni} \propto \sum_{i=1}^N L_i T_i(\tau_{ni} w_i)^{-\theta} = \Psi_n,$$

where the integral evaluates to a constant that depends only on ρ and θ (using $\theta > \rho - 1$). Hence $\tilde{P}_n \propto \Psi_n^{-1/\theta}$.

Finally, to derive the trade share, use the optimal pricing rule (43), quantity demanded, which must equal to the quantity produced, the equilibrium measure of firms $\tilde{M}_{ni} \propto L_i T_i(w_i \tau_{ni})^{-\theta} (\tilde{c}_n)^\theta$, and the Pareto distribution:

$$\frac{\tilde{X}_{ni}}{\tilde{X}_n} = \frac{\tilde{M}_{ni} \int_0^{\tilde{c}_n} p_{ni}(j)^{1-\rho} dF(c_{ni})}{\sum_{k=1}^N \tilde{M}_{nk} \int_0^{\tilde{c}_n} p_{nk}(j)^{1-\rho} dF(c_{nk})} = \frac{L_i T_i(\tau_{ni} w_i)^{-\theta}}{\sum_{k=1}^N L_k T_k(\tau_{nk} w_k)^{-\theta}} = \frac{L_i T_i(\tau_{ni} w_i)^{-\theta}}{\Psi_n}.$$

c. We refer the reader to Arkolakis et al. (2012) for derivations concerning welfare in all the models. ■

The proof of the ranking between the price gap distributions of the EK and BEJK models at the boundary points follows.

Proof: We need to show that $P_{\mathcal{EK}}(\tilde{P}_h = \log \tau) > P_{\mathcal{BEJK}}(\tilde{P}_h = \log \tau)$ and $P_{\mathcal{EK}}(\tilde{P}_h = -\log \tau) > P_{\mathcal{BEJK}}(\tilde{P}_h = -\log \tau)$. Let $\tilde{P}_{h,\mathcal{EK}}$ and $\tilde{P}_{h,\mathcal{BEJK}}$ denote the logged price gaps in the EK and BEJK models respectively. Following the notation in the text, $c_{khf}, k = 1, 2$ denotes the k th best supplier from F to H. We show that $\tilde{P}_{h,\mathcal{EK}} = \log \tau$ holds true in higher number of instances than $\tilde{P}_{h,\mathcal{BEJK}} = \log \tau$, and similarly for the lower bound, $-\log \tau$.

Broadly, there are two cases to consider: (1) traded goods (either exported by the home or foreign country) and (2) non-traded goods. For each case, there are sub cases that correspond to different price gaps in the BEJK model as pricing is based on the next best competitor. In the EK model, price gaps always reflect cost gaps due to perfect competition.

1. Traded goods

- (a) Foreign (F) exports to Home (H). Then F is the lowest cost supplier in both F and H. There are three sub cases:

- i. F is the second lowest cost supplier in both F and H. Then,

$$\begin{aligned}\tilde{P}_{h,\mathcal{EK}} &= \log c_{1hf} - \log c_{1ff} = \log \tau \\ \tilde{P}_{h,\mathcal{BEJK}} &= \log c_{2hf} - \log c_{2ff} = \log \tau \\ \Rightarrow \tilde{P}_{h,\mathcal{EK}} &= \tilde{P}_{h,\mathcal{BEJK}} = \log \tau\end{aligned}$$

- ii. F is the second lowest cost supplier in F; H is the second lowest cost supplier in H. Then,

$$\begin{aligned}\tilde{P}_{h,\mathcal{EK}} &= \log c_{1hf} - \log c_{1ff} = \log \tau \\ \tilde{P}_{h,\mathcal{BEJK}} &= \log c_{1hh} - \log c_{2ff} < \log \tau \\ \Rightarrow \tilde{P}_{h,\mathcal{EK}} &= \log \tau > \tilde{P}_{h,\mathcal{BEJK}}\end{aligned}$$

- iii. H is the second lowest cost supplier in both F and H. This violates no-arbitrage and cannot occur.

- (b) Home (H) exports to Foreign (F). Then H is the lowest cost supplier in both F and H. There are three sub cases:

- i. H is the second lowest cost supplier in both F and H. Then,

$$\begin{aligned}\tilde{P}_{h,\mathcal{EK}} &= \log c_{1hh} - \log c_{1fh} = -\log \tau \\ \tilde{P}_{h,\mathcal{BEJK}} &= \log c_{2hh} - \log c_{2fh} = -\log \tau \\ \Rightarrow \tilde{P}_{h,\mathcal{EK}} &= \tilde{P}_{h,\mathcal{BEJK}} = -\log \tau\end{aligned}$$

- ii. F is the second lowest cost supplier in F; H is the second lowest cost supplier in H. Then,

$$\begin{aligned}\tilde{P}_{h,\mathcal{EK}} &= \log c_{1hh} - \log c_{1fh} = -\log \tau \\ \tilde{P}_{h,\mathcal{BEJK}} &= \log c_{2hh} - \log c_{1ff} > -\log \tau \\ \Rightarrow \tilde{P}_{h,\mathcal{EK}} &= -\log \tau < \tilde{P}_{h,\mathcal{BEJK}}\end{aligned}$$

- iii. F is the second lowest cost supplier in both F and H. This violates no-arbitrage and cannot occur.

All four sub cases occur with positive probability. Given the strict inequality in sub cases a.ii and b.ii, it must be that $P_{\mathcal{EK}}(\tilde{P}_h = \log \tau) > P_{\mathcal{BEJK}}(\tilde{P}_h = \log \tau)$ and $P_{\mathcal{EK}}(\tilde{P}_h = -\log \tau) > P_{\mathcal{BEJK}}(\tilde{P}_h = -\log \tau)$. There remain four sub cases that correspond to non-traded goods. In each sub case, $P_{h,\mathcal{EK}}, P_{h,\mathcal{BEJK}} \in (-\log \tau, \log \tau)$, but

the two random variables cannot be ordered theoretically. Hence, while the mass that lies in the interior is larger in the BEJK versus the EK model, it cannot be shown analytically how this mass is distributed. It is for this reason that it cannot be shown analytically that $G_{\mathcal{EK}}$ stochastically dominates $G_{\mathcal{BEJK}}$. We present the remaining four sub cases for completeness.

2. Non-traded goods. F is the lowest cost supplier in F and H is the lowest cost supplier in H. There are four sub cases:

- (a) F is the second lowest cost supplier in both F and H. Then,

$$\begin{aligned}\tilde{P}_{h,\mathcal{EK}} &= \log c_{1hh} - \log c_{1ff} \in (-\log \tau, \log \tau) \\ \tilde{P}_{h,\mathcal{BEJK}} &= \log c_{1hf} - \log c_{2ff} = \log \tau + \log c_{1ff} - \log c_{2ff} \in (-\log \tau, \log \tau)\end{aligned}$$

- (b) H is the second lowest cost supplier in both F and H. Then,

$$\begin{aligned}\tilde{P}_{h,\mathcal{EK}} &= \log c_{1hh} - \log c_{1ff} \in (-\log \tau, \log \tau) \\ \tilde{P}_{h,\mathcal{BEJK}} &= \log c_{2hh} - \log c_{1fh} = \log c_{1hh} - \log \tau - \log c_{1hh} \in (-\log \tau, \log \tau)\end{aligned}$$

- (c) F is the second lowest cost supplier in F; H is the second lowest cost supplier in H. Then,

$$\begin{aligned}\tilde{P}_{h,\mathcal{EK}} &= \log c_{1hh} - \log c_{1ff} \in (-\log \tau, \log \tau) \\ \tilde{P}_{h,\mathcal{BEJK}} &= \log c_{2hh} - \log c_{2ff} \in (-\log \tau, \log \tau)\end{aligned}$$

- (d) H is the second lowest cost supplier in F; F is the second lowest cost supplier in H. Then,

$$\begin{aligned}\tilde{P}_{h,\mathcal{EK}} &= \log c_{1hh} - \log c_{1ff} \in (-\log \tau, \log \tau) \\ \tilde{P}_{h,\mathcal{BEJK}} &= \log c_{1hf} - \log c_{1fh} = \log c_{1ff} - \log c_{1hh} \in (-\log \tau, \log \tau)\end{aligned}$$

■

B Performance on Simulated Data

To provide evidence that the estimation procedure recovers the underlying parameter for each model, we apply the methodology on data simulated by each model under a known θ . For exposition purposes, we let θ be the same across the models, and we set it equal to four.

Table 9: Estimation Results With Artificial Data, Underlying $\theta = 4$

	Model	Estimate of θ	Mean d_{ni}	Cov($d_{ni}, \log m_{ni}$)	Mean d_{ni}^{85th}
Exactly Identified	Armington	3.99 [3.83, 4.26]	1.27	—	—
	EK	3.99 [3.80, 4.18]	0.96	—	—
	BEJK	3.99 [3.84, 4.15]	0.63	—	—
Over Identified	Armington	3.97 [3.82, 4.19]	1.21	0.20	0.24
	EK	3.99 [3.90, 4.10]	0.96	0.10	0.37
	BEJK	4.00 [3.88, 4.10]	0.63	0.04	0.25

Note: Value is the median estimate across simulations. Values within brackets report 90th and 10th percentiles. In each simulation there are 30 countries and 100 simulations are performed. Over Identified uses the optimal weighting matrix; using the identity matrix yielded near identical results.

The first column in Table 9 reports the results. Our estimation routine recovers the value of θ in all three models, both in the exactly identified and overidentified case. Foreshadowing our empirical findings, the final three columns in Table 9 report the average moments across simulations. Consistent with the theoretical results in Section 4, there is a systematic ranking in largest price gaps with Armington being the highest and BEJK being the lowest. Consistent with Proposition 2, this implies that given observed price gaps, the models will be ordered by θ to match this data.

C Robustness Results

C.1 BEJK

Tables 10 and 11 present the estimates for the BEJK model under three alternative assumed values for ρ ranging between 1.5 to 3. The estimated θ increases monotonically with ρ . Interestingly, at lower levels of ρ , the data no longer reject the BEJK model in year 2004. Therefore, it appears that the richest model comes closest to matching the price moments in the data when the trade elasticity falls below 3.

Table 10: BEJK Sensitivity to ρ : Over-Identified

ρ	Year	θ	(p10–p90)	J -stat	J pctl
1.5	2004	2.76	2.71 – 2.83	23.3	0.68
1.5	2011	2.80	2.74 – 2.88	173.2	1.00
1.5	2017	2.88	2.81 – 2.95	180.7	1.00
2.0	2004	2.96	2.91 – 3.03	61.6	0.97
2.0	2011	2.97	2.92 – 3.05	180.1	1.00
2.0	2017	3.07	3.00 – 3.14	174.3	1.00
3.0	2004	3.26	3.21 – 3.34	93.3	0.99
3.0	2011	3.25	3.19 – 3.34	195.1	0.99
3.0	2017	3.37	3.31 – 3.45	189.5	1.00

Table 11: BEJK Sensitivity to ρ : Exactly Identified

ρ	Year	θ	(p10–p90)
1.5	2004	2.82	2.73 – 2.93
1.5	2011	2.73	2.66 – 2.85
1.5	2017	2.87	2.76 – 3.00
2.0	2004	3.07	2.95 – 3.17
2.0	2011	2.99	2.89 – 3.09
2.0	2017	3.12	3.00 – 3.22
3.0	2004	3.43	3.28 – 3.53
3.0	2011	3.33	3.22 – 3.45
3.0	2017	3.49	3.36 – 3.61

D Data

Table 12 lists the 64 traded basic heading categories from the ICP used in the analysis.

Table 12: Traded Basic Heading Categories

#	Basic Heading	#	Basic Heading
1	Rice	33	Tobacco
2	Other cereals, flour, cereal prod.	34	Clothing materials and accessories
3	Bread	35	Garments
4	Other bakery products	36	Shoes and other footwear
5	Pasta products and couscous	37	Furniture and furnishings
6	Beef and veal	38	Carpets and floor coverings
7	Pork	39	Household textiles
8	Lamb, mutton and goat	40	Major household appliances
9	Poultry	41	Small electric household appliances
10	Other meats and preparations	42	Glassware, tableware, utensils
11	Fresh, chilled or frozen fish	43	Major tools and equipment
12	Preserved or processed fish	44	Small tools and misc. accessories
13	Fresh milk	45	Non-durable household goods
14	Preserved milk and milk prod.	46	Pharmaceutical products
15	Cheese and curd	47	Other medical products
16	Eggs and egg-based products	48	Therapeutic appliances and equip.
17	Butter and margarine	49	Motor cars
18	Other edible oils and fats	50	Motor cycles
19	Fresh or chilled fruit	51	Bicycles
20	Frozen, preserved fruit prod.	52	Fuels and lubricants, personal transport
21	Fresh vegetables (excl. potatoes)	53	Telephone and telefax equipment
22	Potatoes and tuber vegetables	54	Audio-visual, photo, IT equipment
23	Frozen, preserved vegetable prod.	55	Recording media
24	Sugar	56	Other recreational items and equip.
25	Jams, marmalades and honey	57	Newspapers, books and stationery
26	Confectionery, chocolate, ice cream	58	Personal care products
27	Food products n.e.c.	59	Jewellery, clocks and watches
28	Coffee, tea and cocoa	60	Fabricated metal products
29	Mineral waters, soft drinks, juices	61	Electrical and optical equipment
30	Spirits	62	General purpose machinery
31	Wine	63	Special purpose machinery
32	Beer	64	Road transport equipment

Table 13 reports summary statistics for the 85th percentile price gap statistic, $d_{ni2,t}^d$.

Table 13: Summary Statistics: $d_{ni2,t}^d$ (85th Percentile Price Gap)

Year	N	Basic Headings	Mean	Median	Std	[Min, Max]
All	2604	—	0.343	0.344	0.126	[0.076, 0.773]
2004	866	62	0.369	0.380	0.119	[0.100, 0.765]
2011	868	71	0.338	0.335	0.128	[0.076, 0.735]
2017	870	64	0.323	0.310	0.125	[0.081, 0.773]

The d_{ni2} measure is substantially smaller (mean of 0.34 versus 0.94) and less dispersed, as expected from using the 85th percentile instead of the maximum. There is a slight downward trend across years, which is consistent with declining trade barriers over time. Table 14 reports the results of the Pearson correlations.

Table 14: Correlations: $d_{ni2,t}^d$ and Gravity Variables

Year	N	$\rho(d_{ni2}, \log \text{dist})$	p	$\rho(d_{ni2}, \text{border})$	p	$\rho(d_{ni2}, \log(1 + \tau))$
All	2604	0.501	0.000	-0.208	0.000	0.349
2004	866	0.371	0.000	-0.144	0.000	0.246
2011	868	0.605	0.000	-0.263	0.000	0.404
2017	870	0.536	0.000	-0.222	0.000	0.418

Notes: All p -values are below 10^{-4} .

The $d_{ni2,t}^d$ measure exhibits slightly stronger distance correlations than $d_{ni,t}^d$, suggesting that the 85th percentile may be less noisy than the maximum. Table 15 reports the regression results, which are similar to the results in the main text.

Table 15: OLS Regression: $d_{ni2,t}^d$

Variable	Coef.	Std. Err.	t	p	95% CI
Intercept	-0.084	0.020	-4.25	0.000	[-0.122, -0.045]
Border	-0.001	0.010	-0.11	0.909	[-0.020, 0.018]
$\log(\text{dist})$	0.055	0.003	21.53	0.000	[0.050, 0.060]
$\log(1 + \tau)$	0.394	0.057	6.85	0.000	[0.281, 0.506]
$R^2 = 0.287, \quad N = 2,604, \quad F = 209.6$					